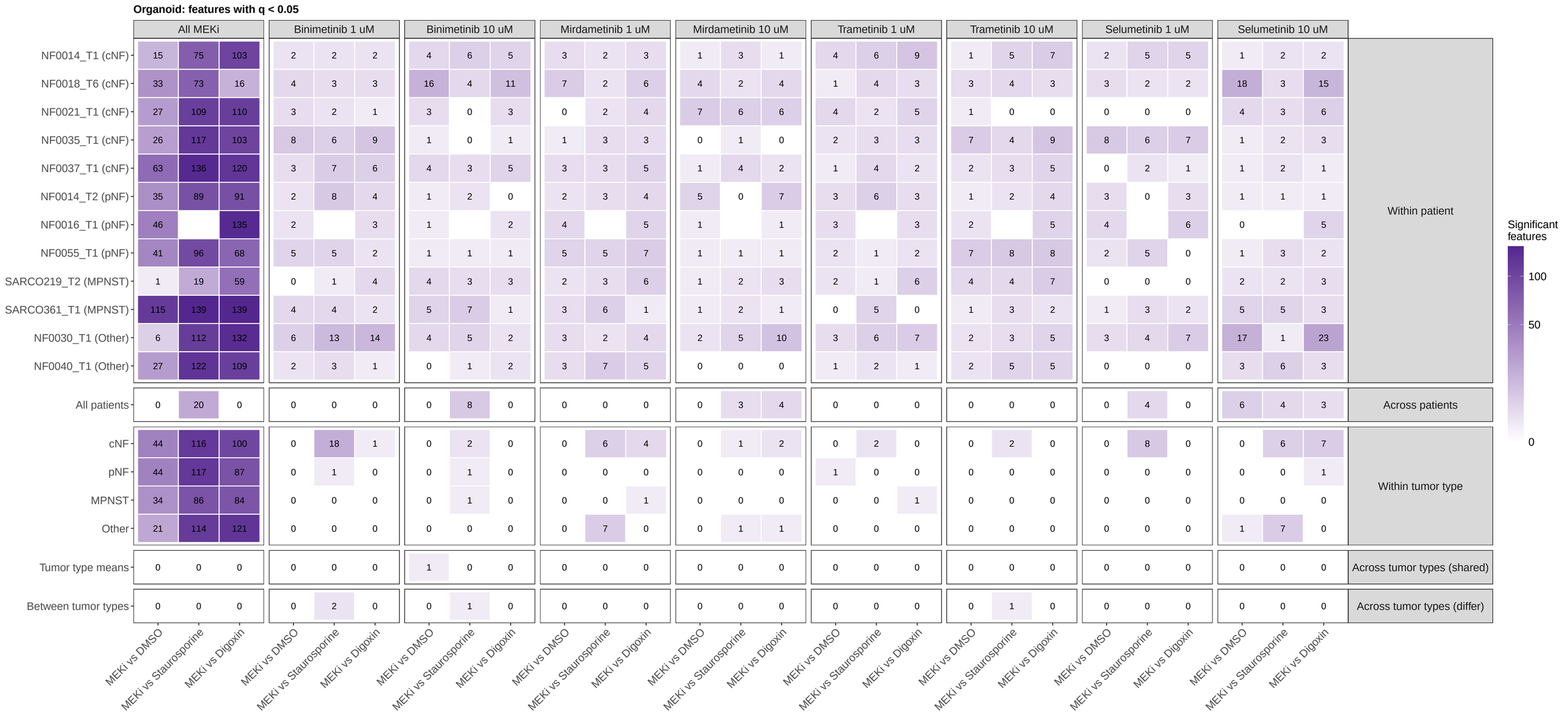
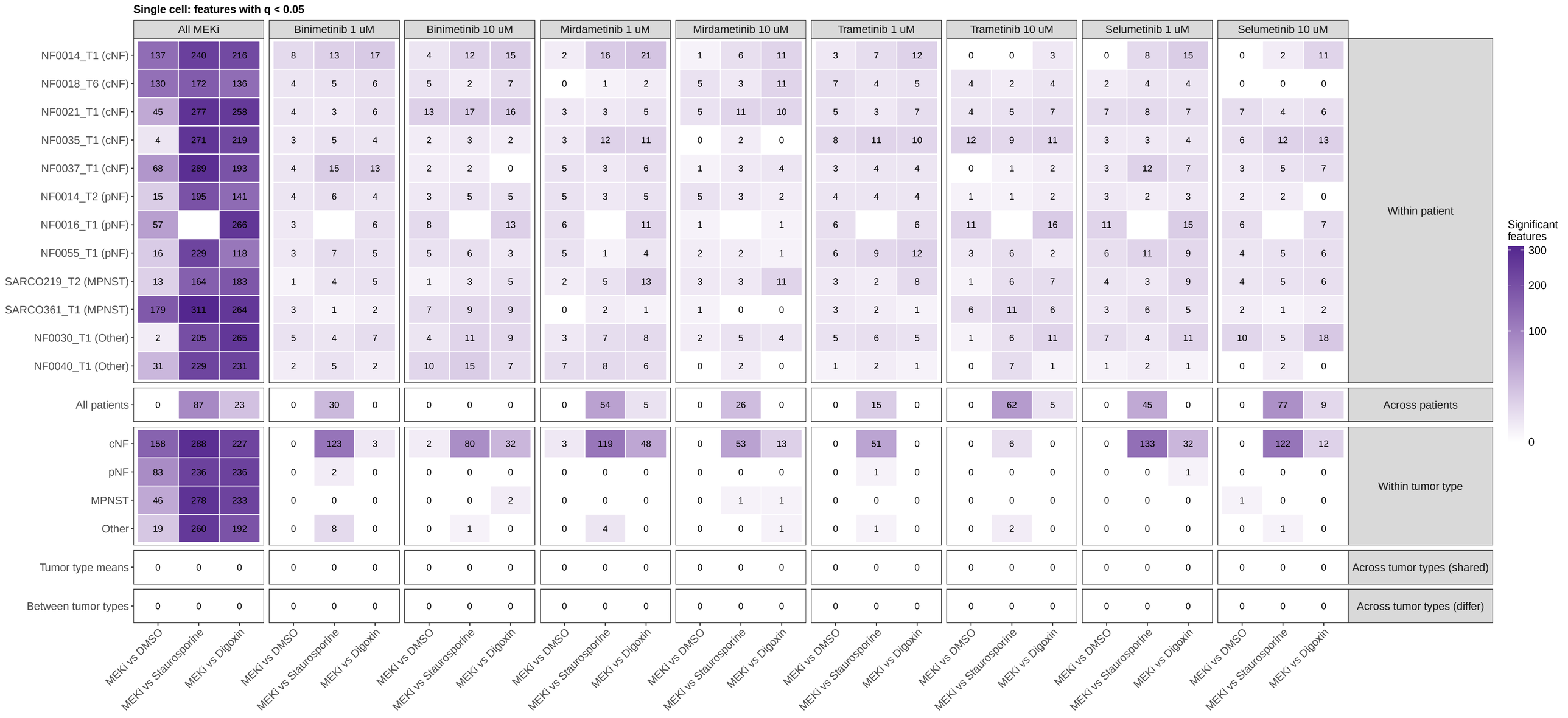
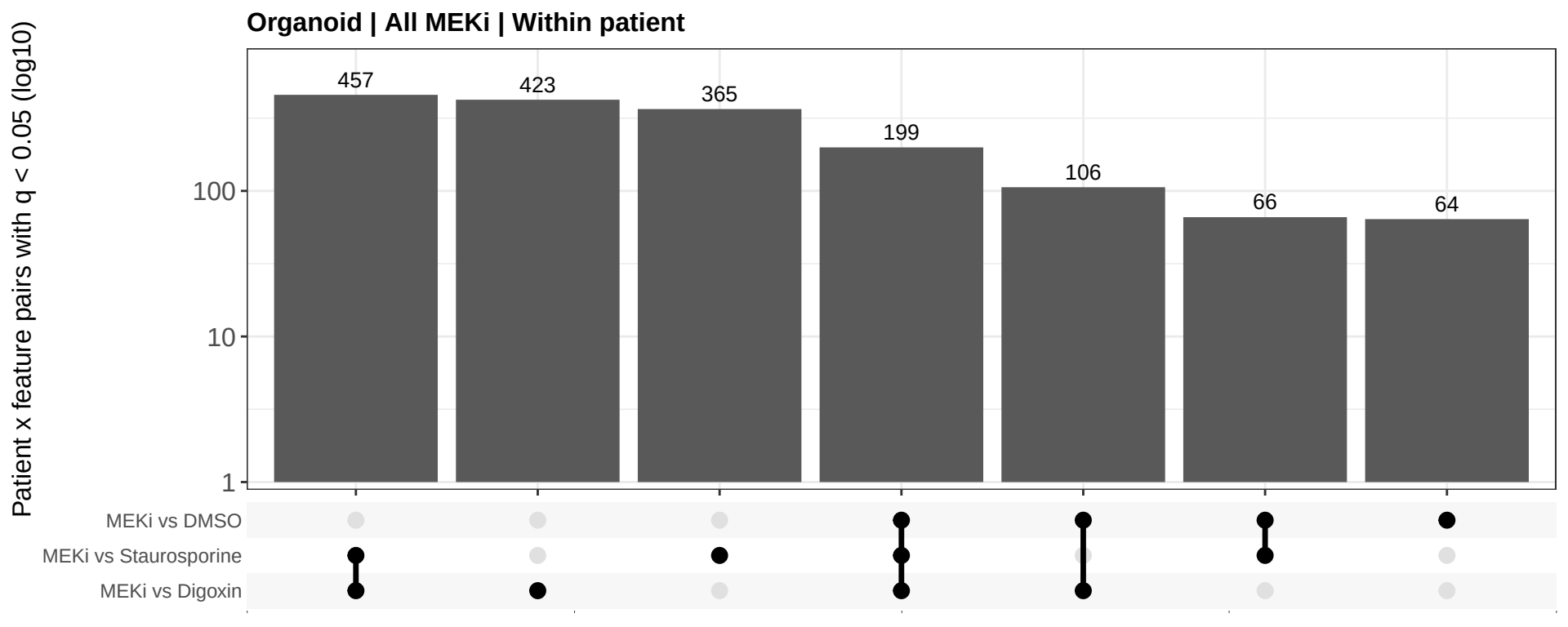
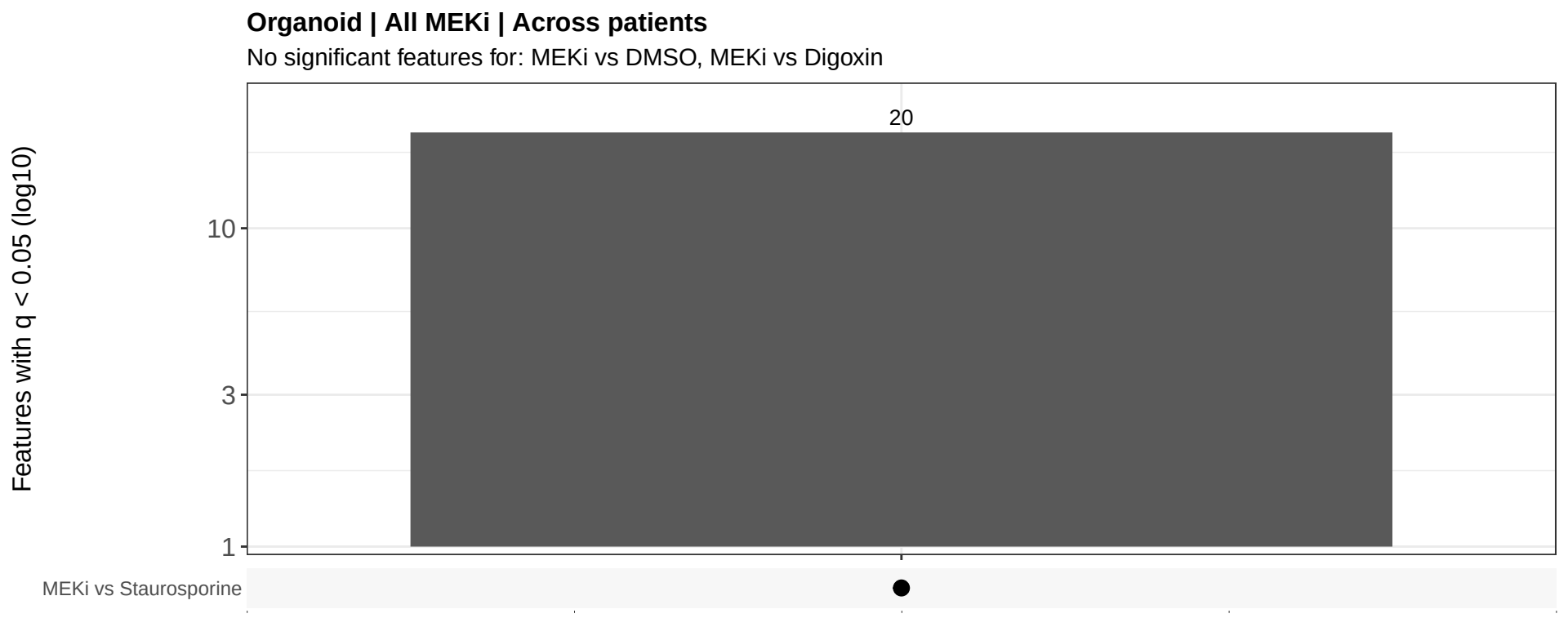
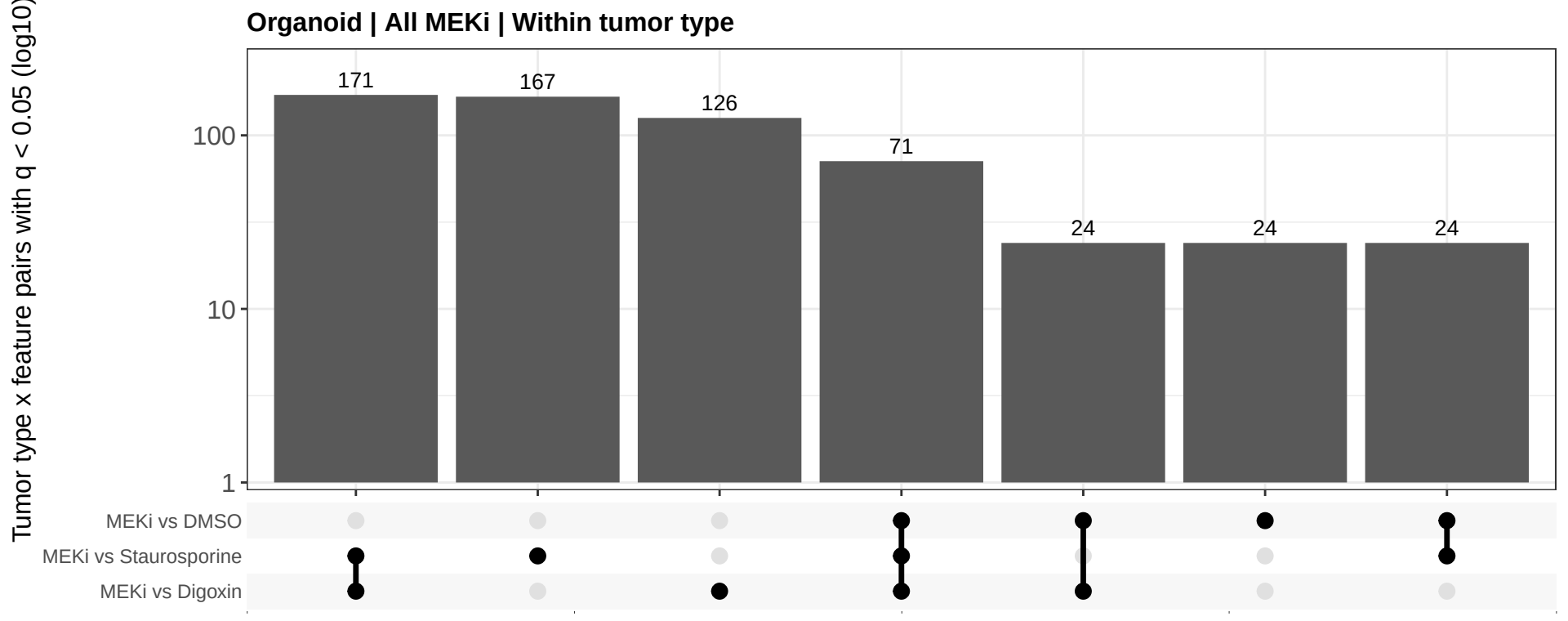

Organoid: features with q < 0.05
All MEKi
Binimetinib 1 uM
Binimetinib 10 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Trametinib 1 uM
Trametinib 10 uM
Selumetinib 1 uM
Selumetinib 10 uM
NF0014_T1 (cNF) 15 75 103 2 2 2 4 6 5 3 2 3 1 3 1 4 6 9 1 5 7 2 5 5 1 2 2
NF0018_T6 (cNF) 33 73 16 4 3 3 16 4 11 7 2 6 4 2 4 1 4 3 3 4 3 3 2 2 18 3 15
NF0021_T1 (cNF) 27 109 110 3 2 1 3 0 3 0 2 4 7 6 6 4 2 5 1 0 0 0 0 0 4 3 6
NF0035_T1 (cNF) 26 117 103 8 6 9 1 0 1 1 3 3 0 1 0 2 3 3 7 4 9 8 6 7 1 2 3
NF0037_T1 (cNF) 63 136 120 3 7 6 4 3 5 3 3 5 1 4 2 1 4 2 2 3 5 0 2 1 1 2 1
NF0014_T2 (pNF) 35 89 91 2 8 4 1 2 0 2 3 4 5 0 7 3 6 3 1 2 4 3 0 3 1 1 1
NF0016_T1 (pNF) 46 135 2 3 1 2 4 5 1 1 3 3 2 5 4 6 0 5
NF0055_T1 (pNF) 41 96 68 5 5 2 1 1 1 5 5 7 1 1 1 2 1 2 7 8 8 2 5 0 1 3 2
SARCO219_T2 (MPNST) 1 19 59 0 1 4 4 3 3 2 3 6 1 2 3 2 1 6 4 4 7 0 0 0 2 2 3
SARCO361_T1 (MPNST) 115 139 139 4 4 2 5 7 1 3 6 1 1 2 1 0 5 0 1 3 2 1 3 2 5 5 3
NF0030_T1 (Other) 6 112 132 6 13 14 4 5 2 3 2 4 2 5 10 3 6 7 2 3 5 3 4 7 17 1 23
NF0040_T1 (Other) 27 122 109 2 3 1 0 1 2 3 7 5 0 0 0 1 2 1 2 5 5 0 0 0 3 6 3
All patients 0 20 0 0 0 0 0 8 0 0 0 0 0 3 4 0 0 0 0 0 0 0 4 0 6 4 3
cNF 44 116 100 0 18 1 0 2 0 0 6 4 0 1 2 0 2 0 0 2 0 0 8 0 0 6 7
pNF 44 117 87 0 1 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1
MPNST 34 86 84 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
Other 21 114 121 0 0 0 0 0 0 0 7 0 0 1 1 0 0 0 0 0 0 0 0 0 1 7 0
Tumor type means 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Between tumor types 0 0 0 0 2 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
Within patient
Across patients
Within tumor type
Across tumor types (shared)
Across tumor types (differ)
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin
Significant features
100
50
0


Single cell: features with q < 0.05
All MEKi
Binimetinib 1 uM
Binimetinib 10 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Trametinib 1 uM
Trametinib 10 uM
Selumetinib 1 uM
Selumetinib 10 uM
NF0014_T1 (cNF) 137 240 216 8 13 17 4 12 15 2 16 21 1 6 11 3 7 12 0 0 3 0 8 15 0 2 11
NF0018_T6 (cNF) 130 172 136 4 5 6 5 2 7 0 1 2 5 3 11 7 4 5 4 2 4 2 4 4 0 0 0
NF0021_T1 (cNF) 45 277 258 4 3 6 13 17 16 3 3 5 5 11 10 5 3 7 4 5 7 7 8 7 7 4 6
NF0035_T1 (cNF) 4 271 219 3 5 4 2 3 2 3 12 11 0 2 0 8 11 10 12 9 11 3 3 4 6 12 13
NF0037_T1 (cNF) 68 289 193 4 15 13 2 2 0 5 3 6 1 3 4 3 4 4 0 1 2 3 12 7 3 5 7
NF0014_T2 (pNF) 15 195 141 4 6 4 3 5 5 5 3 5 5 3 2 4 4 4 1 1 2 3 2 3 2 2 0
NF0016_T1 (pNF) 57 266 3 6 8 13 6 11 1 1 6 6 11 16 11 15 6 7
NF0055_T1 (pNF) 16 229 118 3 7 5 5 6 3 5 1 4 2 2 1 6 9 12 3 6 2 6 11 9 4 5 6
SARCO219_T2 (MPNST) 13 164 183 1 4 5 1 3 5 2 5 13 3 3 11 3 2 8 1 6 7 4 3 9 4 5 6
SARCO361_T1 (MPNST) 179 311 264 3 1 2 7 9 9 0 2 1 1 0 0 3 2 1 6 11 6 3 6 5 2 1 2
NF0030_T1 (Other) 2 205 265 5 4 7 4 11 9 3 7 8 2 5 4 5 6 5 1 6 11 7 4 11 10 5 18
NF0040_T1 (Other) 31 229 231 2 5 2 10 15 7 7 8 6 0 2 0 1 2 1 0 7 1 1 2 1 0 2 0
Within patient
All patients 0 87 23 0 30 0 0 0 0 0 54 5 0 26 0 0 15 0 0 62 5 0 45 0 0 77 9
Across patients
cNF 158 288 227 0 123 3 2 80 32 3 119 48 0 53 13 0 51 0 0 6 0 0 133 32 0 122 12
pNF 83 236 236 0 2 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0
MPNST 46 278 233 0 0 0 0 0 2 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 1 0 0
Other 19 260 192 0 8 0 0 1 0 0 4 0 0 0 1 0 1 0 0 2 0 0 0 0 0 1 0
Within tumor type
Tumor type means 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Across tumor types (shared)
Between tumor types 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Across tumor types (differ)
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin
Significant features
300
200
100
0


Organoid | All MEKi | Within patient
Patient x feature pairs with q < 0.05 (log10)
457
423
365
199
106
66
64
100
10
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | All MEKi | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
20
10
3
1
MEKi vs Staurosporine

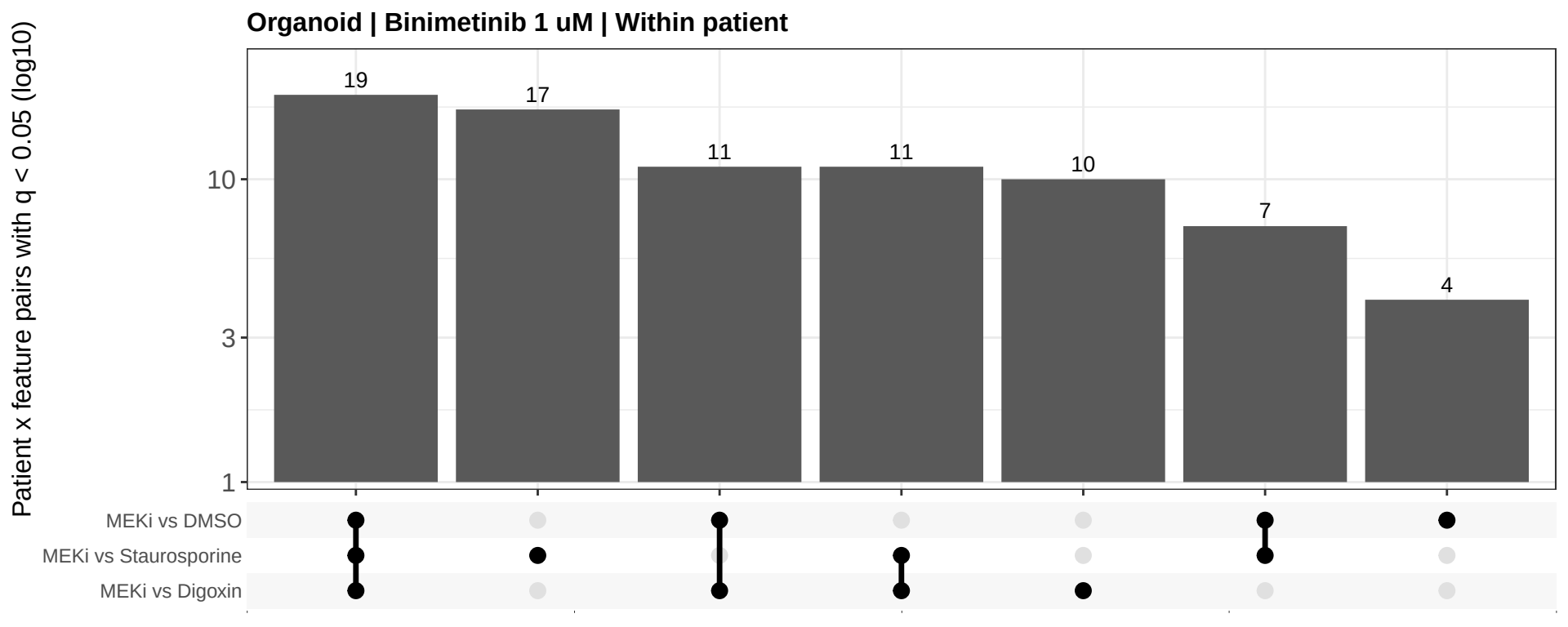

Organoid | All MEKi | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
171
167
126
71
24
24
24
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | All MEKi | Across tumor types (shared)

No features with q < 0.05

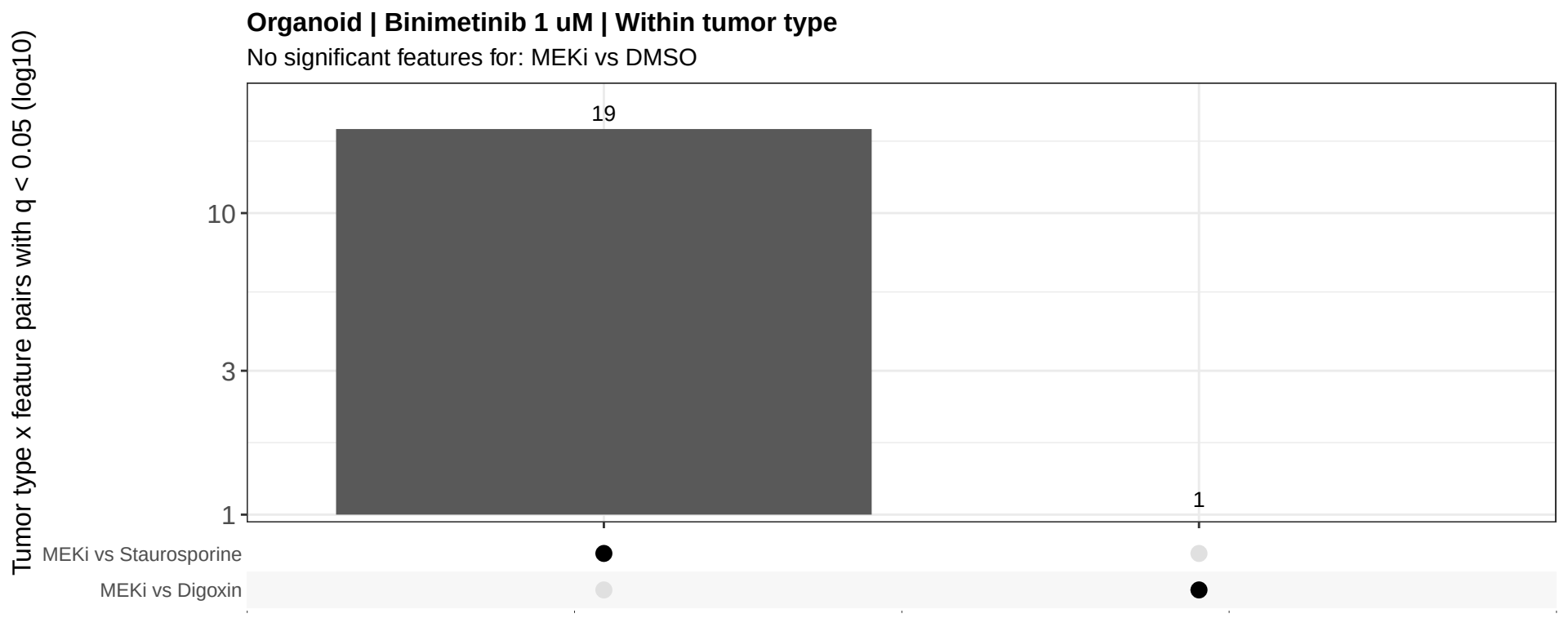
Organoid | All MEKi | Across tumor types (differ)

No features with q < 0.05


Organoid | Binimetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
19
17
11
11
10
7
4
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

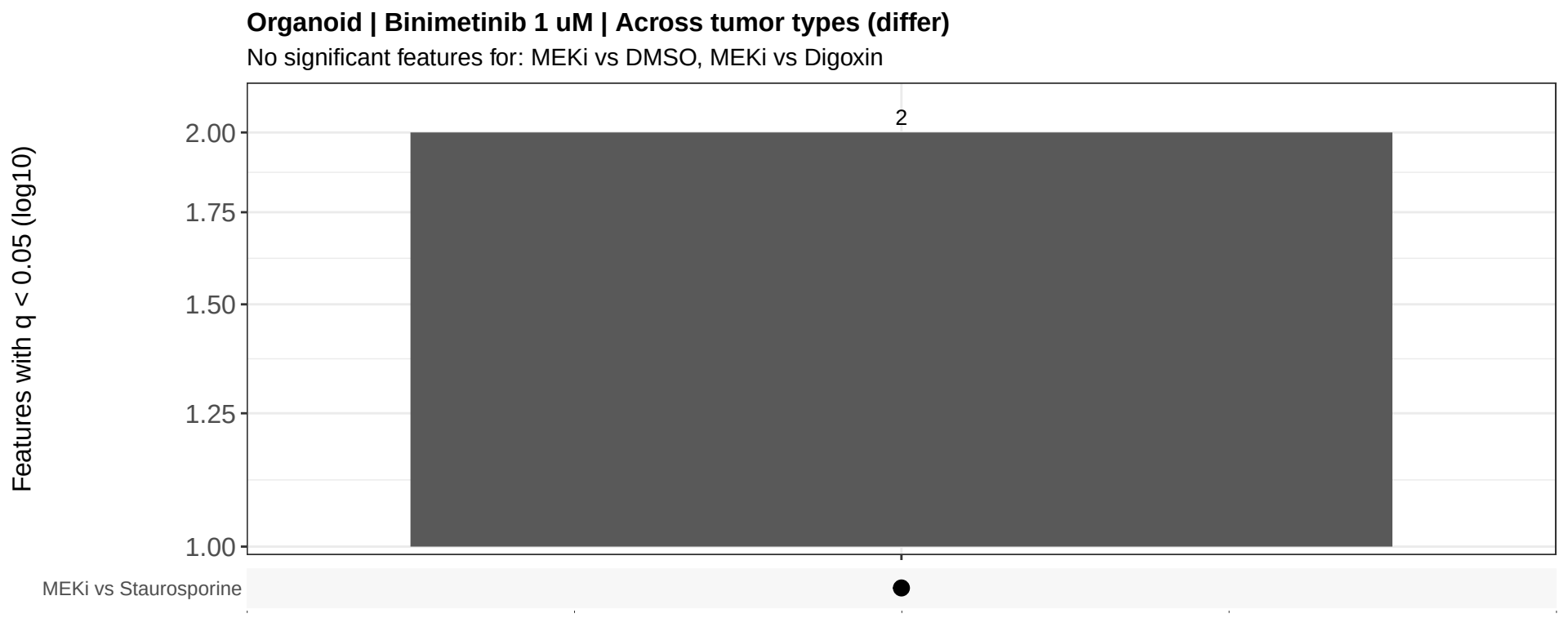
Organoid | Binimetinib 1 uM | Across patients

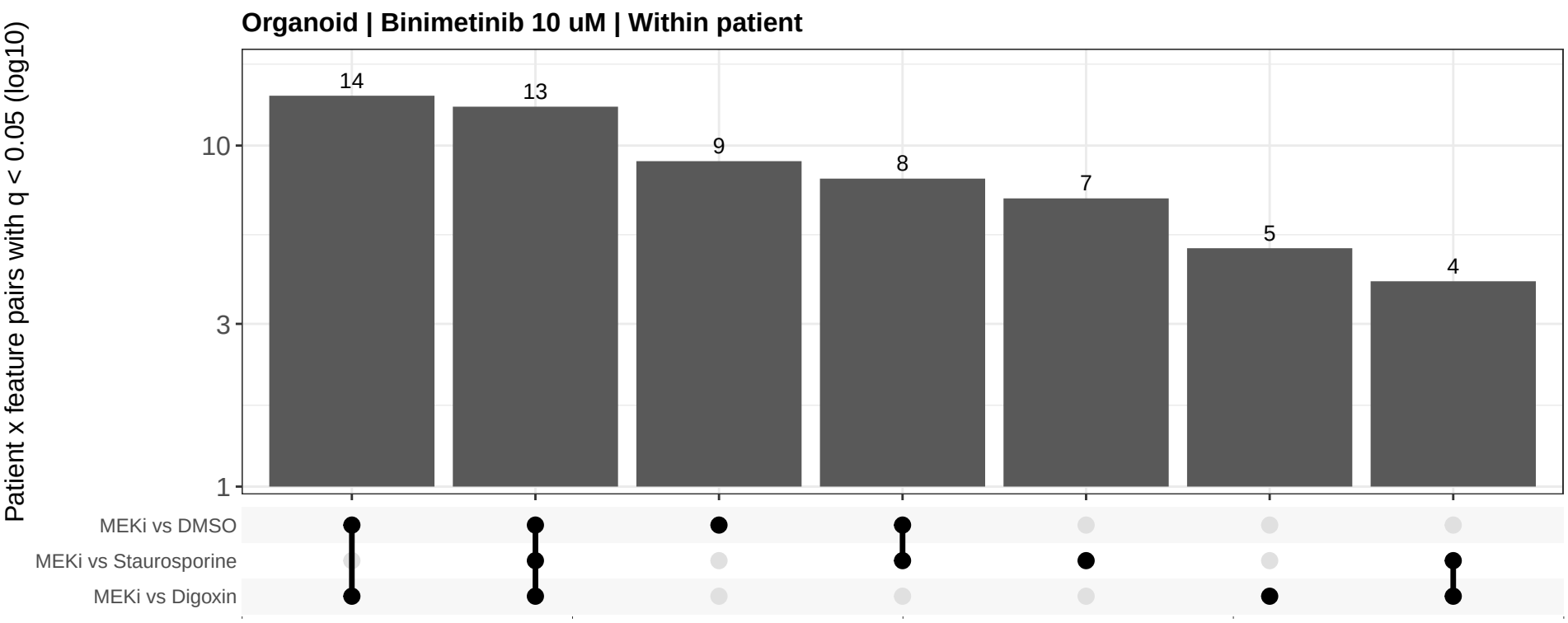
No features with q < 0.05

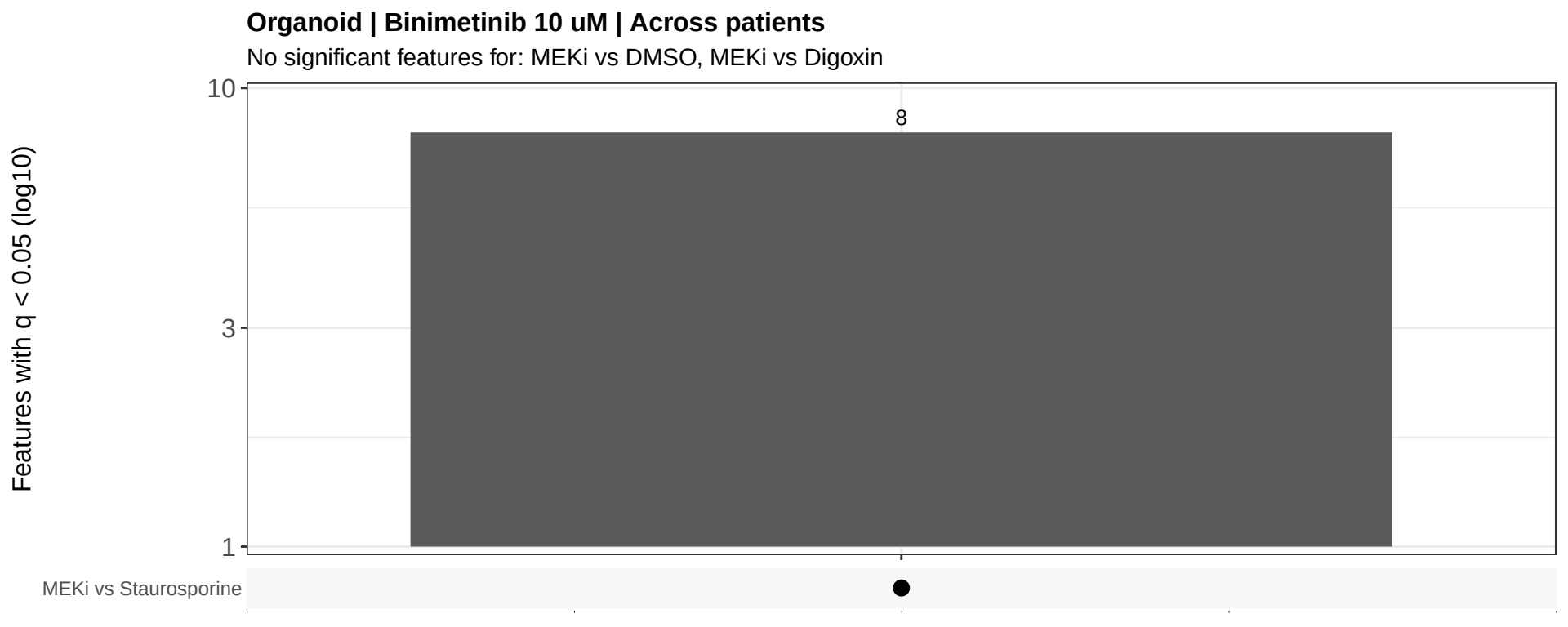

Organoid | Binimetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
19
1
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

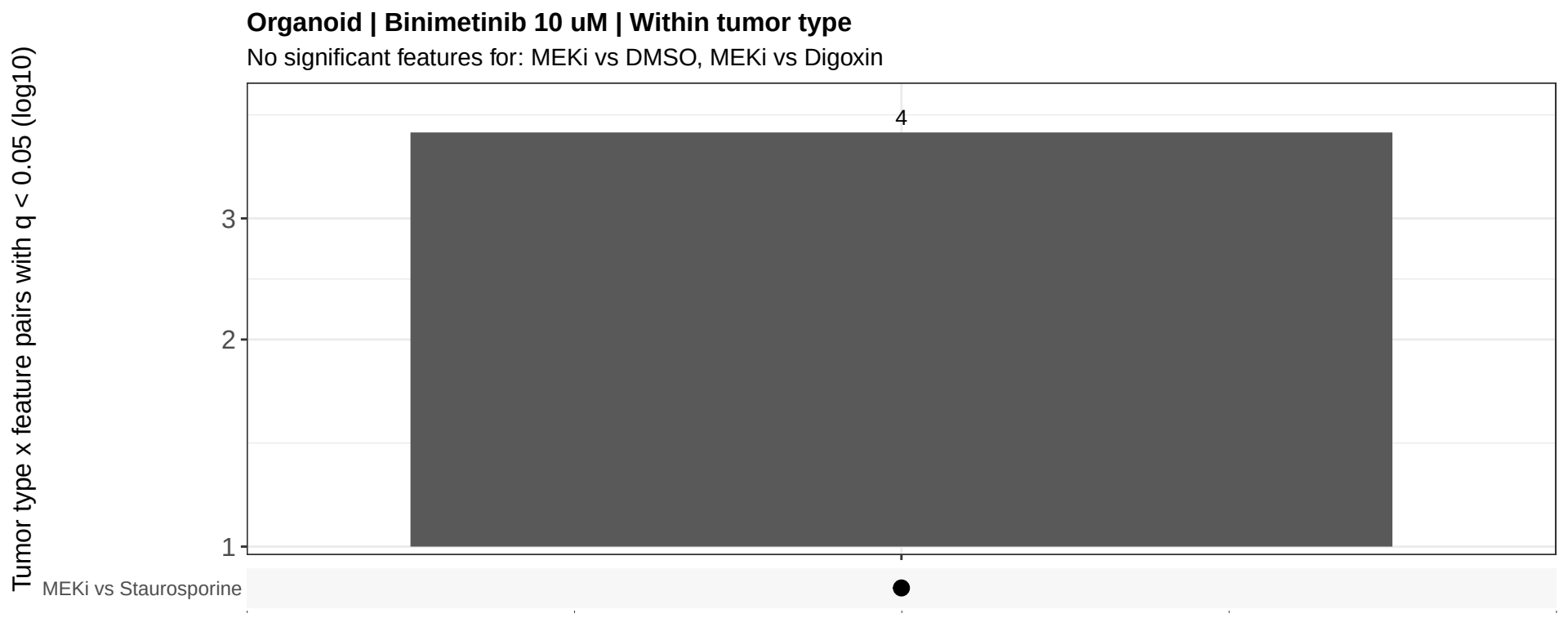
Organoid | Binimetinib 1 uM | Across tumor types (shared)

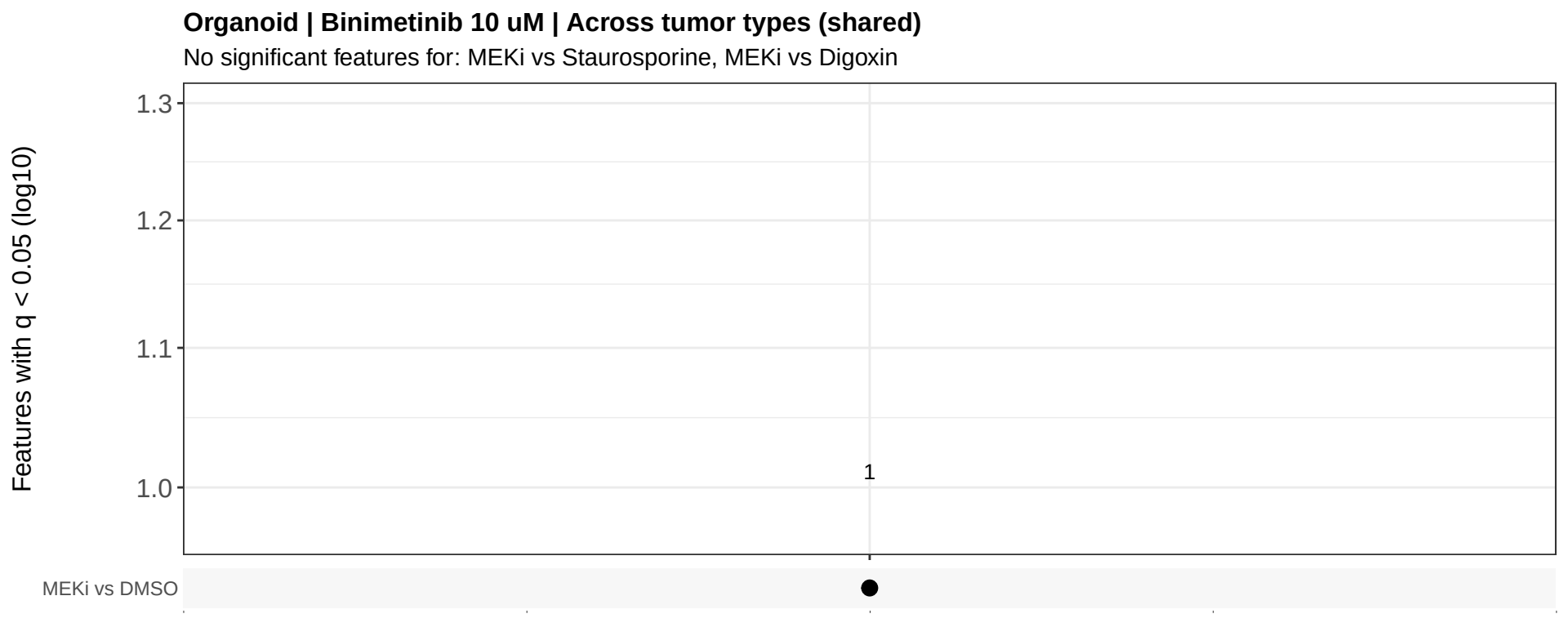
No features with q < 0.05

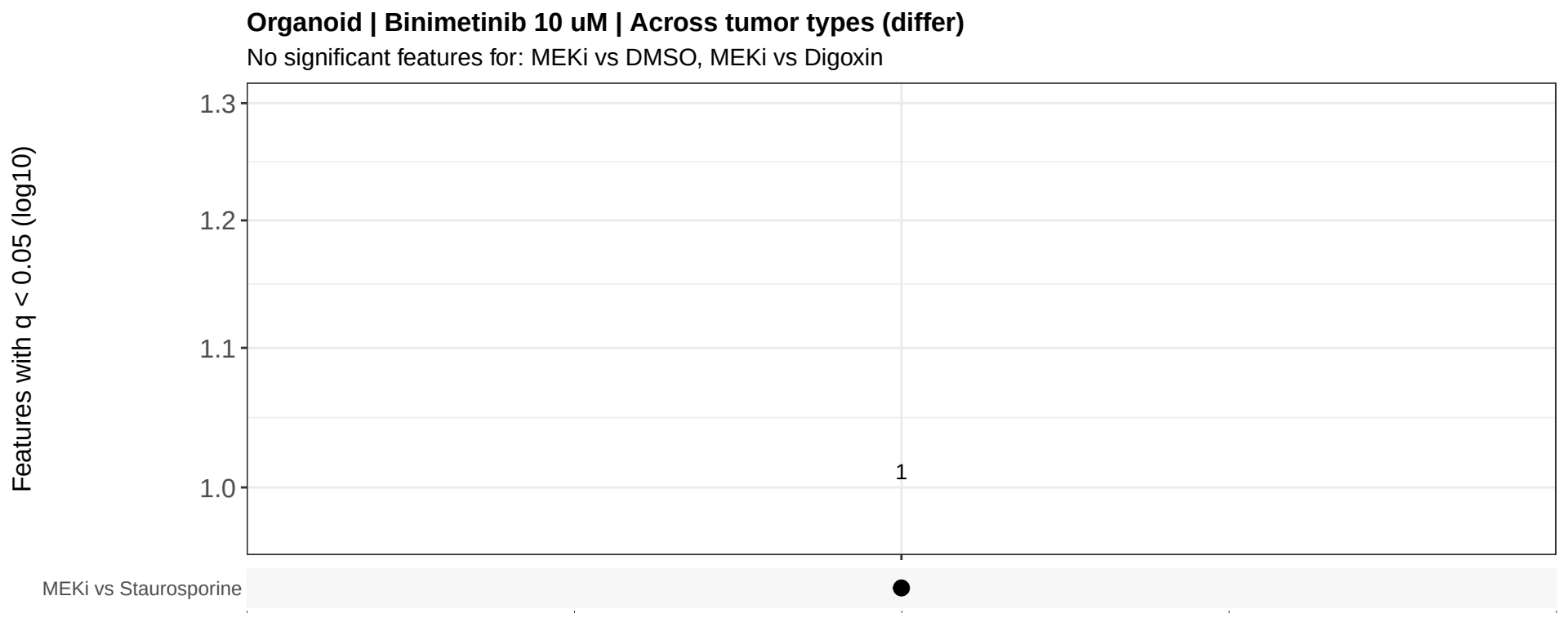

Organoid | Binimetinib 1 uM | Across tumor types (differ)
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
MEKi vs Staurosporine

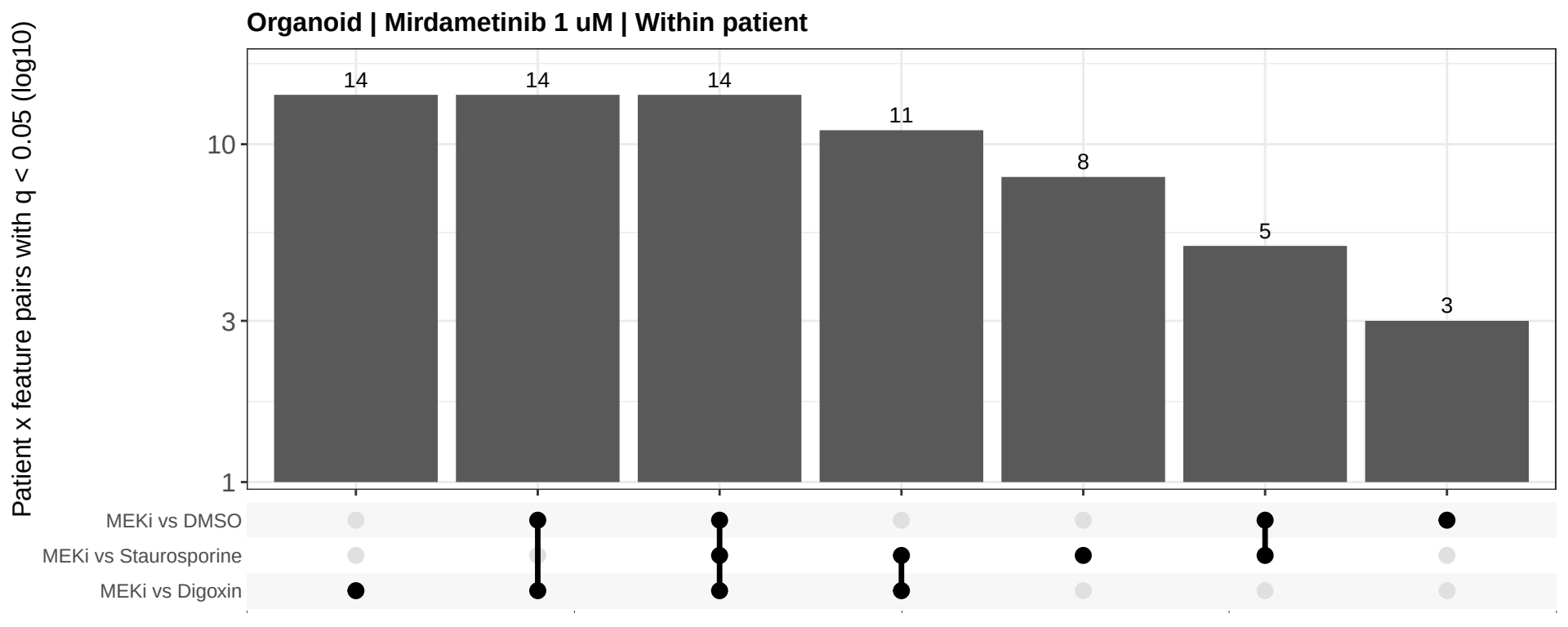

Organoid | Binimetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
10
3
1
14
13
9
8
7
5
4
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Binimetinib 10 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
10
3
1
8
MEKi vs Staurosporine


Organoid | Binimetinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
4
3
2
1
MEKi vs Staurosporine

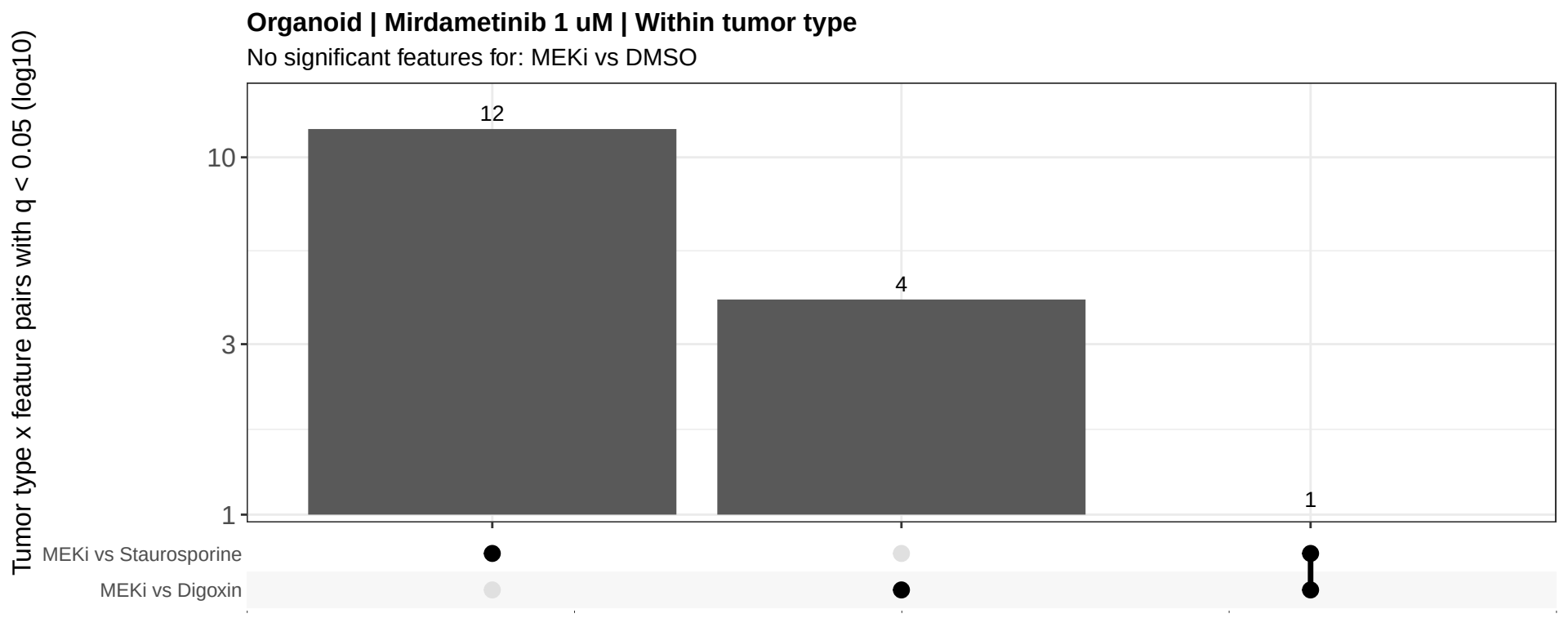

Organoid | Binimetinib 10 uM | Across tumor types (shared)
No significant features for: MEKi vs Staurosporine, MEKi vs Digoxin
Features with q < 0.05 (log10)
1.3
1.2
1.1
1.0
1
MEKi vs DMSO


Organoid | Binimetinib 10 uM | Across tumor types (differ)
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
1.3
1.2
1.1
1.0
1
MEKi vs Staurosporine


Organoid | Mirdametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
14
14
14
11
8
5
3
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Mirdametinib 1 uM | Across patients

No features with q < 0.05

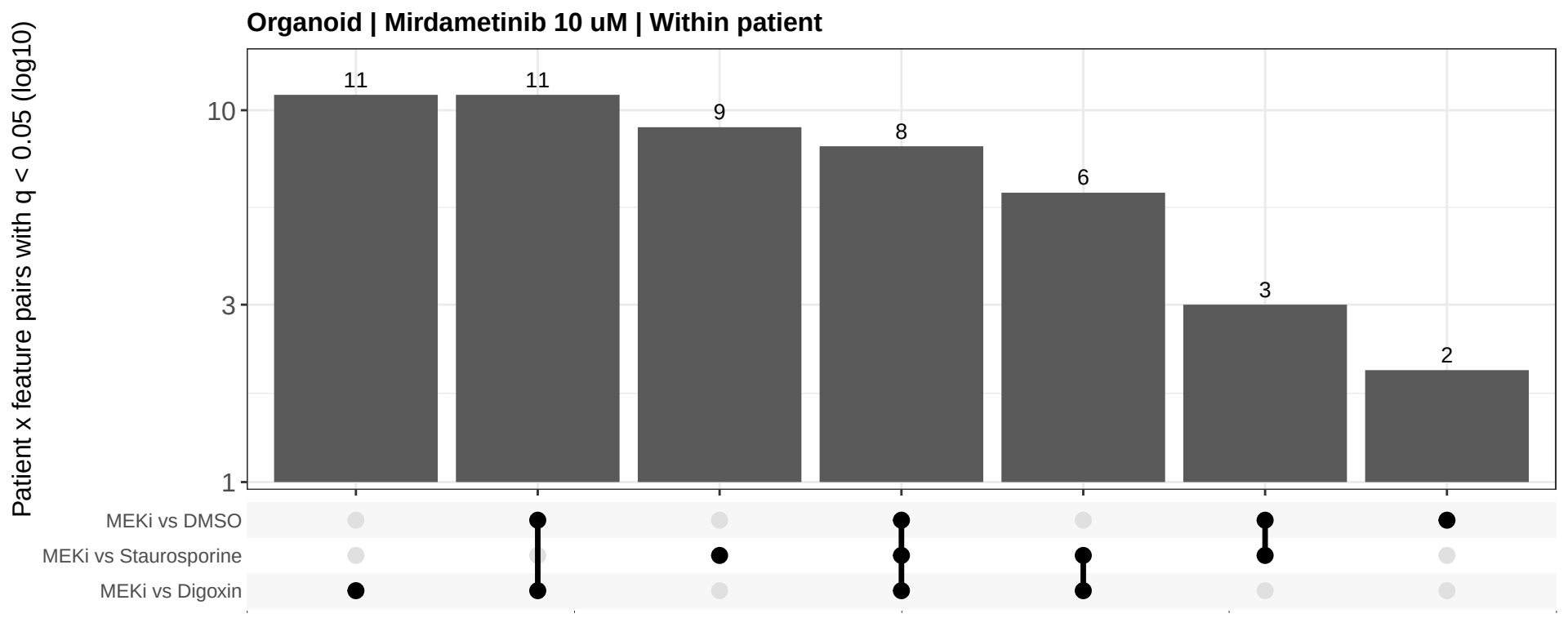

Organoid | Mirdametinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
12
4
1
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

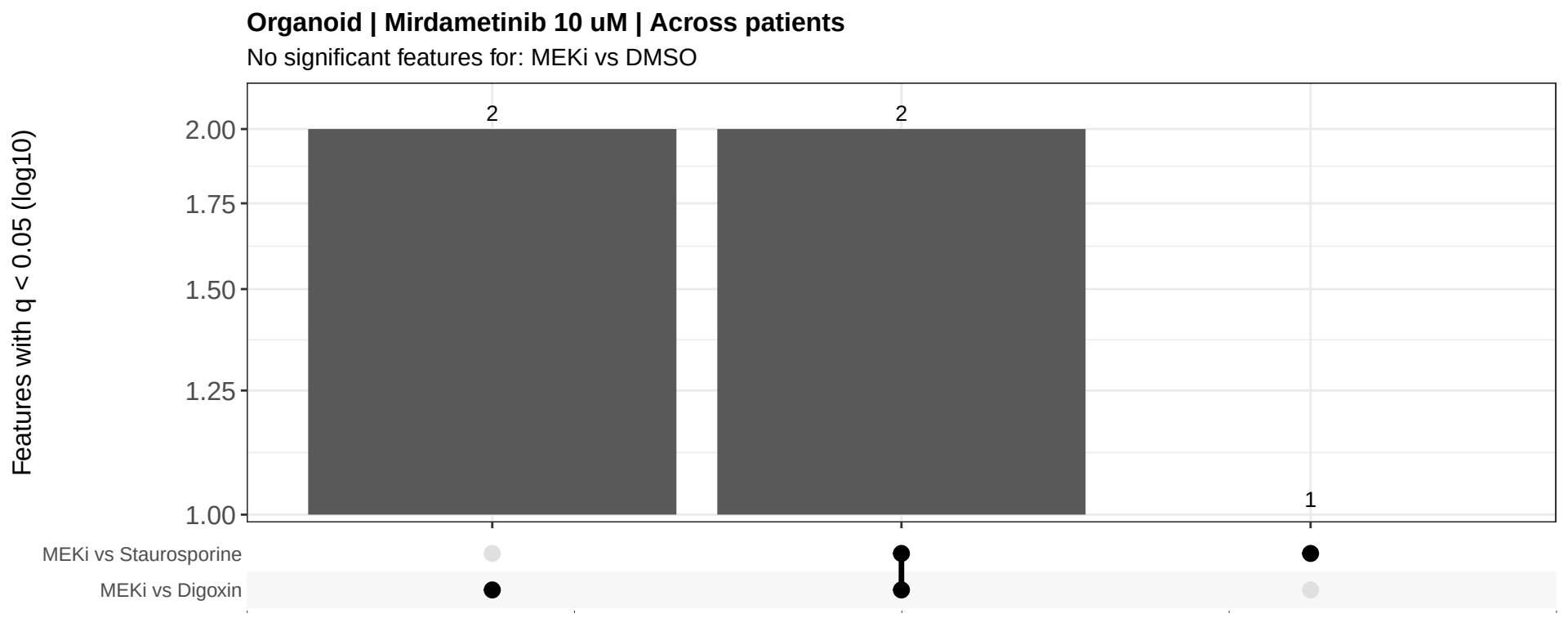
Organoid | Mirdametinib 1 uM | Across tumor types (shared)

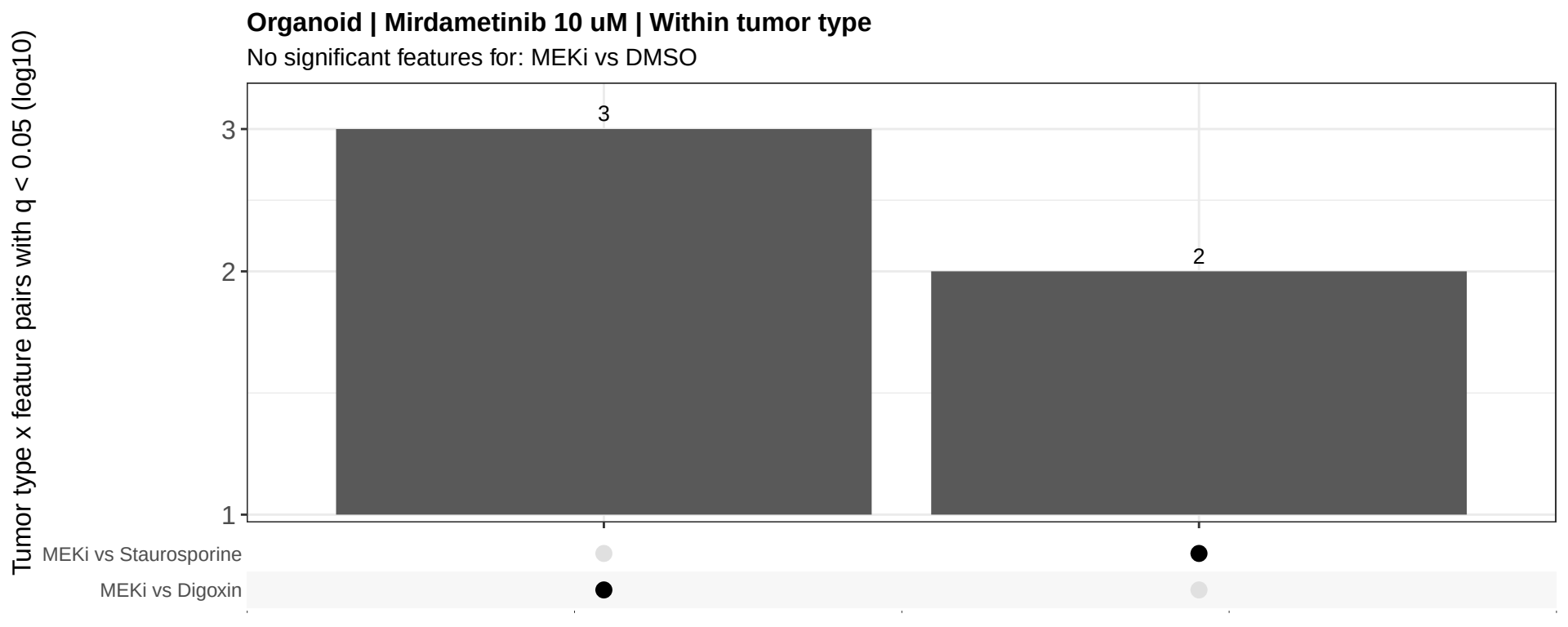
No features with q < 0.05

Organoid | Mirdametinib 1 uM | Across tumor types (differ)

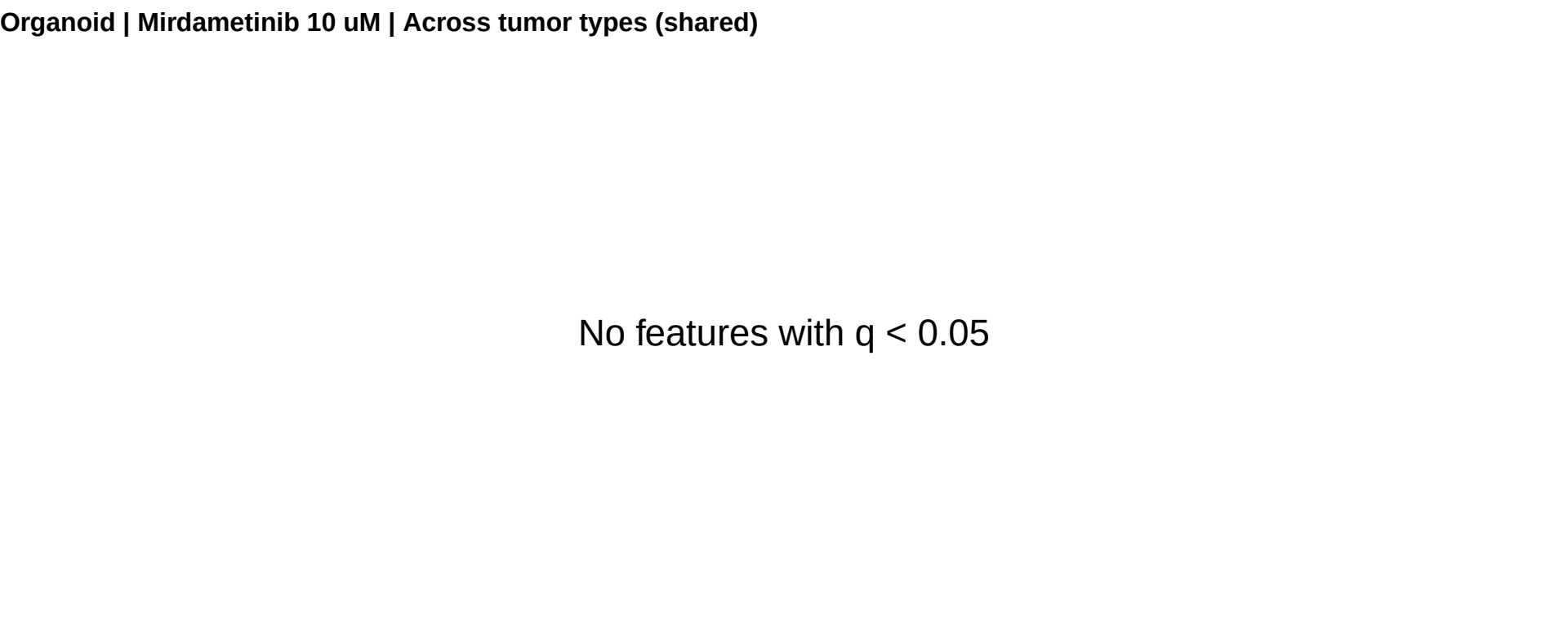
No features with q < 0.05


Organoid | Mirdametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
10
3
1
11
11
9
8
6
3
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Mirdametinib 10 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
2
1
MEKi vs Staurosporine
MEKi vs Digoxin

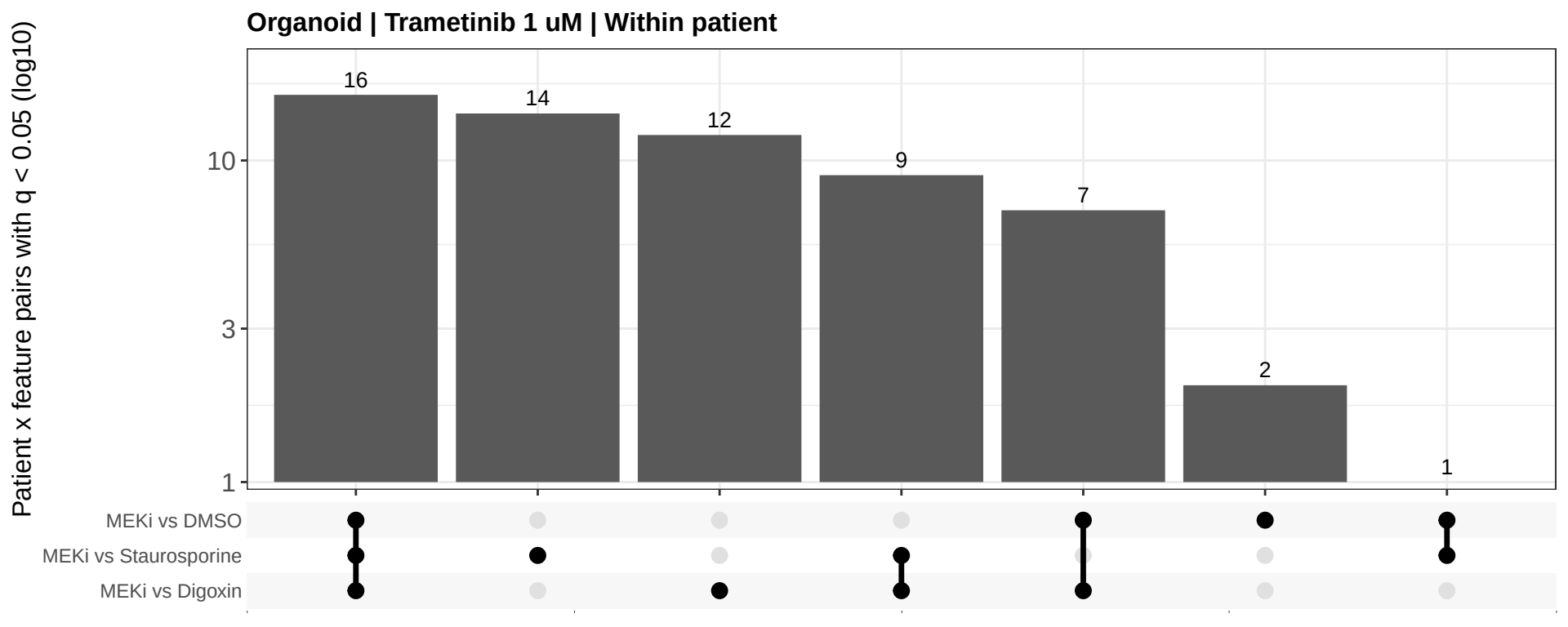

Organoid | Mirdametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
3
2
1
3
2
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Mirdametinib 10 uM | Across tumor types (shared)



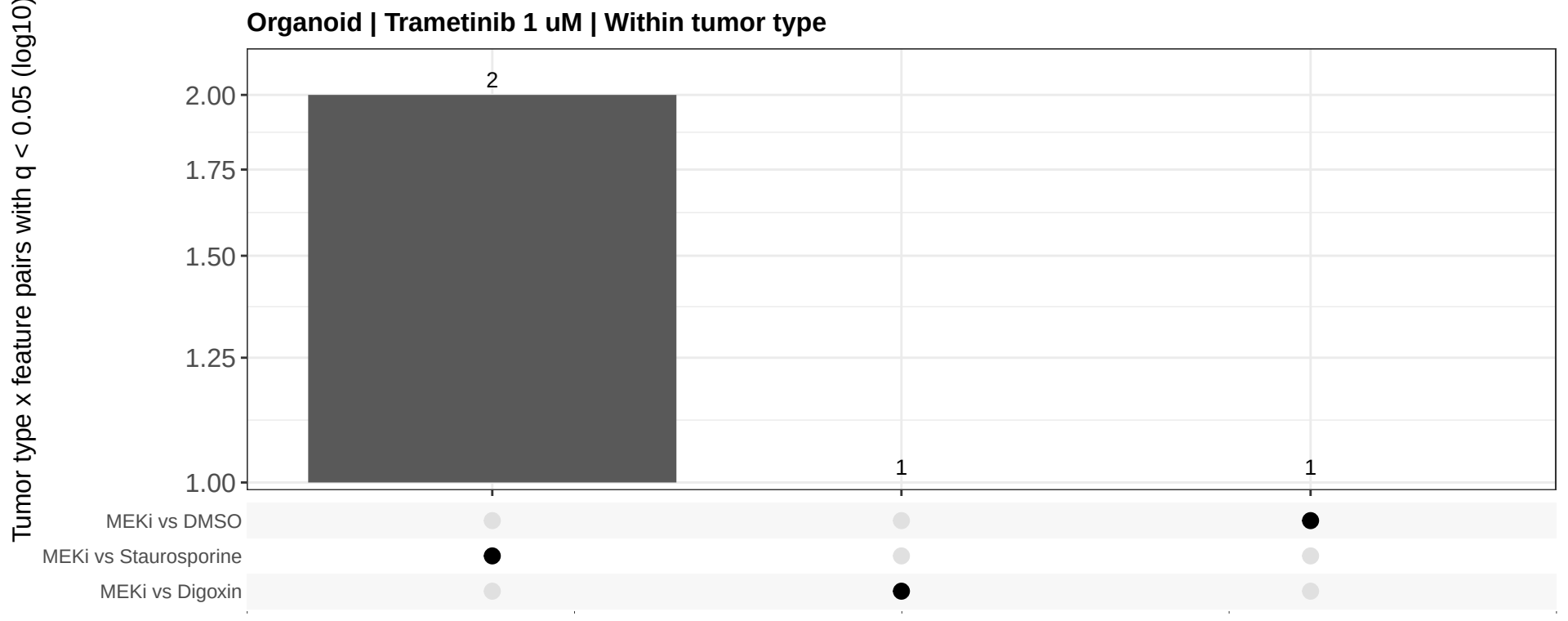
Organoid | Mirdametinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Organoid | Trametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
16
14
12
9
7
2
1
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Trametinib 1 uM | Across patients

No features with q < 0.05

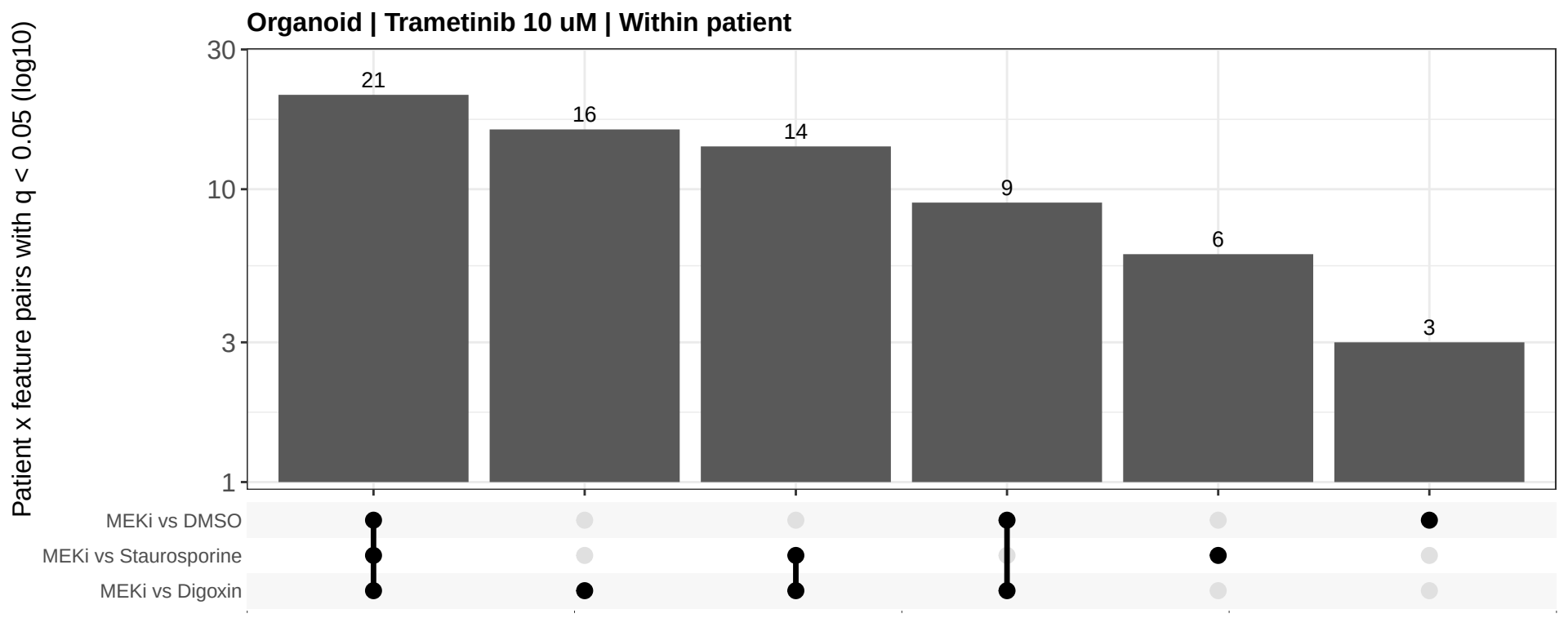

Organoid | Trametinib 1 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
1
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Trametinib 1 uM | Across tumor types (shared)

No features with q < 0.05

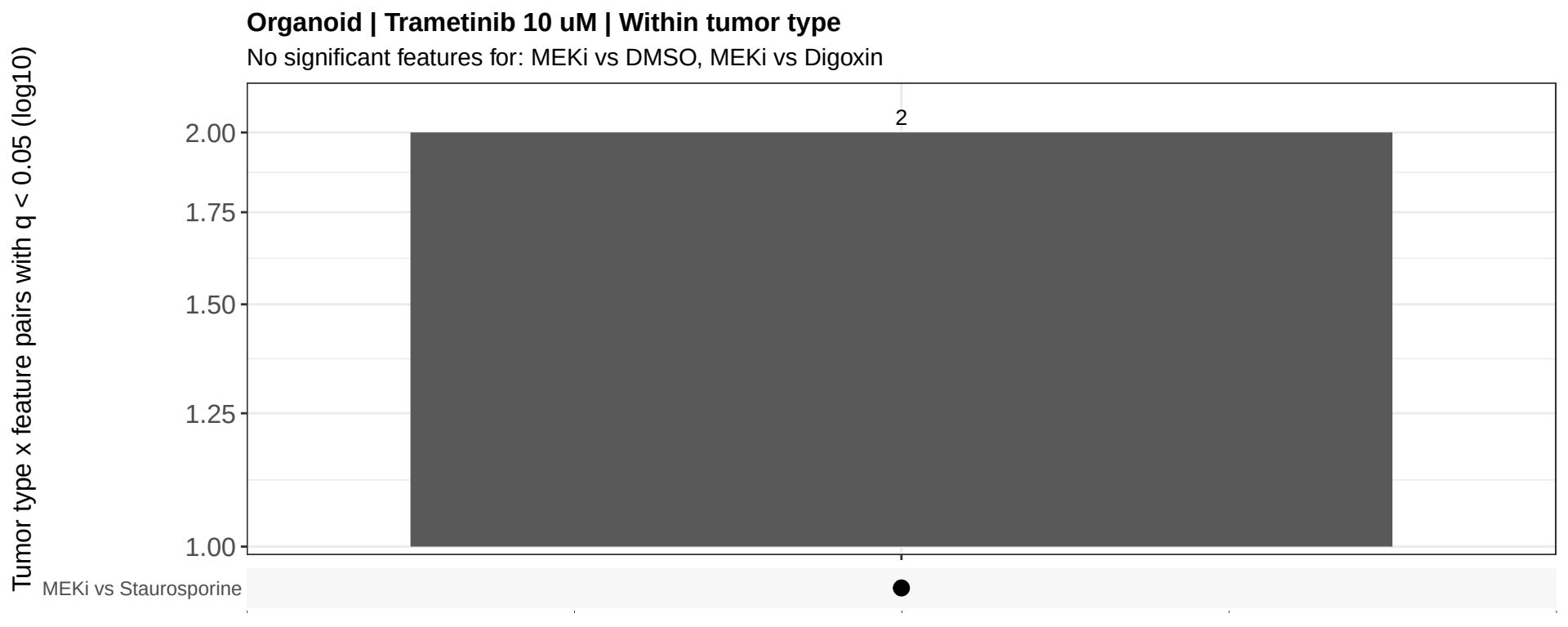
Organoid | Trametinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Organoid | Trametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
21
16
14
9
6
3
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Trametinib 10 uM | Across patients

No features with q < 0.05

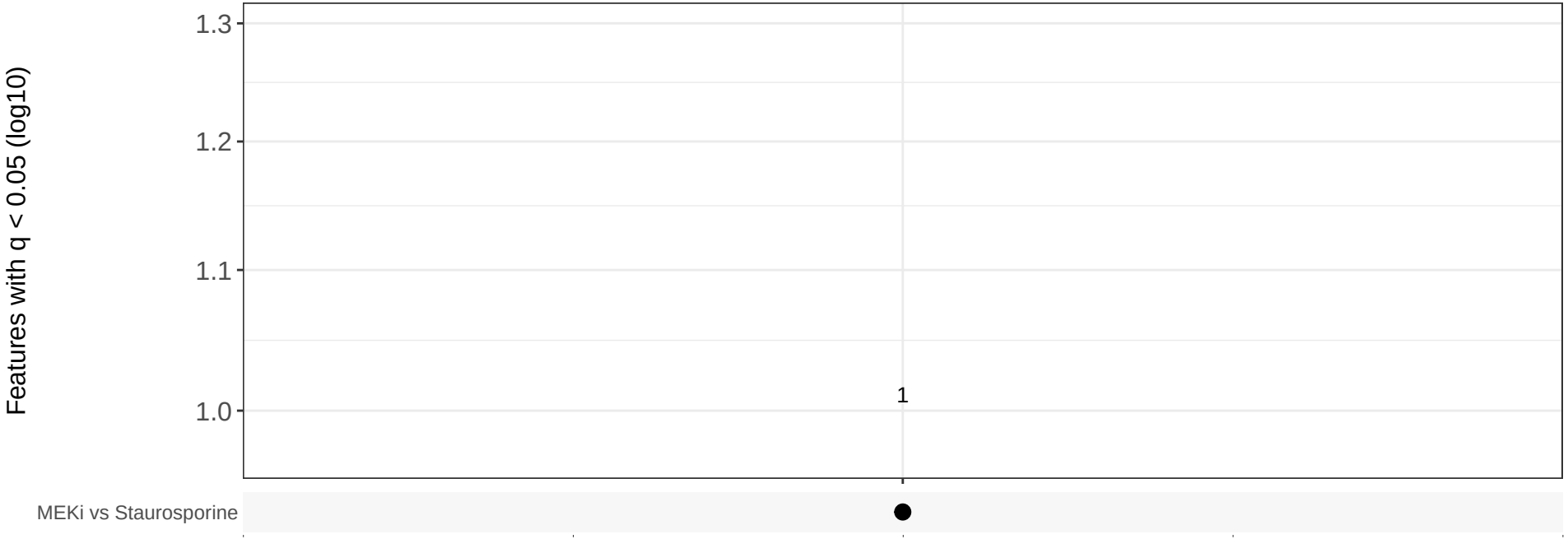

Organoid | Trametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
MEKi vs Staurosporine

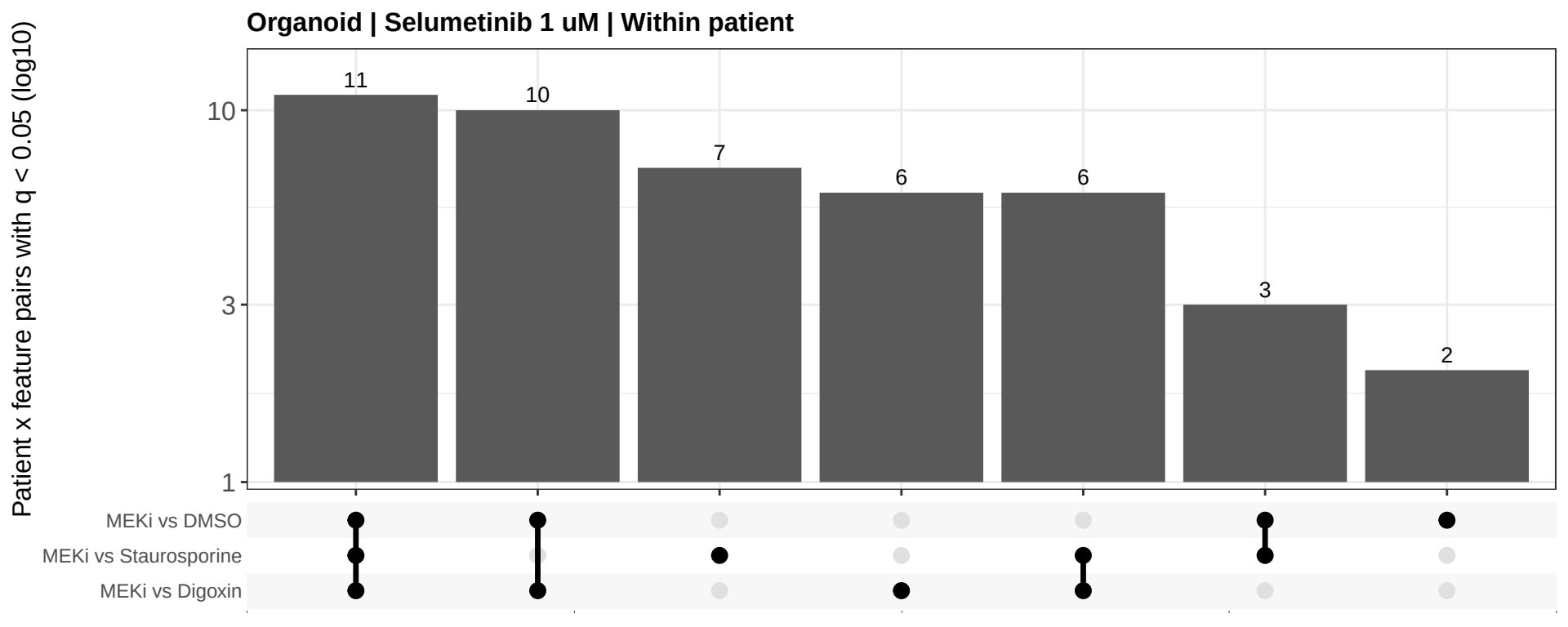
Organoid | Trametinib 10 uM | Across tumor types (shared)

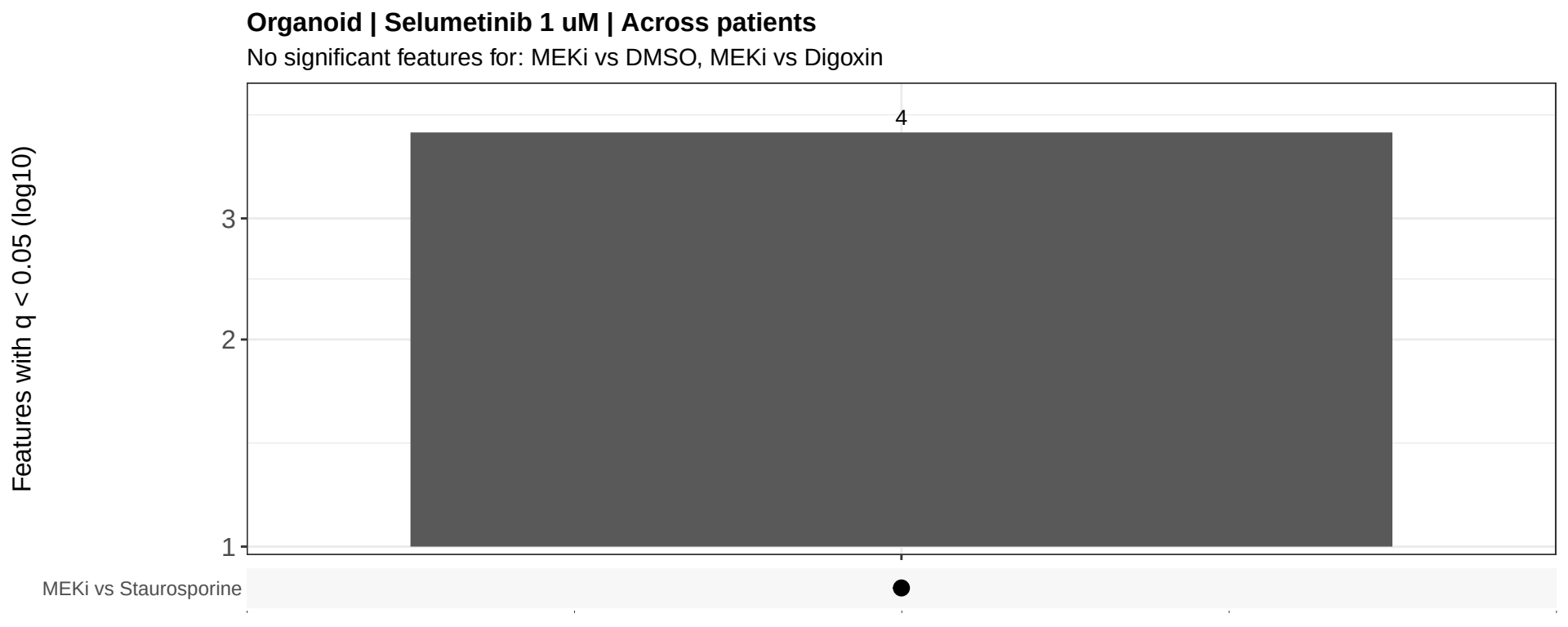
No features with q < 0.05

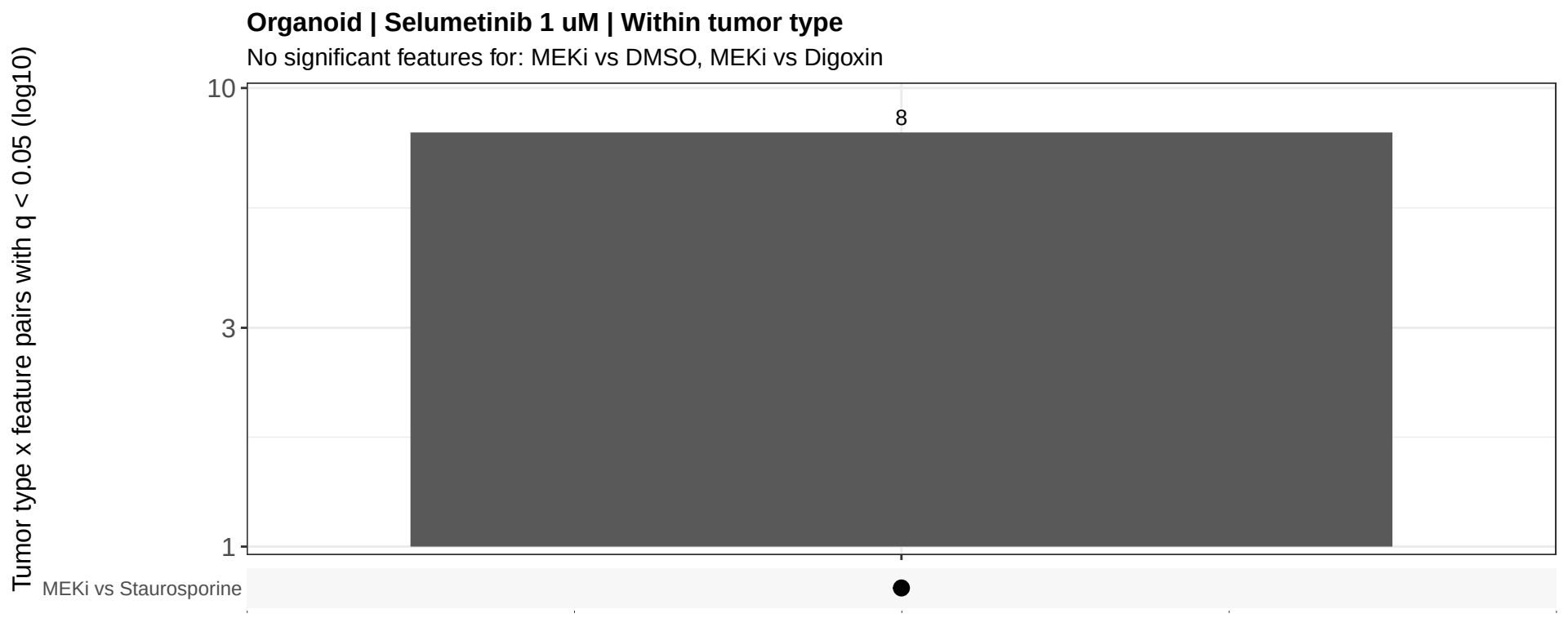
Organoid | Trametinib 10 uM | Across tumor types (differ)

No significant features for: MEKi vs DMSO, MEKi vs Digoxin




Organoid | Selumetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
11
10
7
6
6
3
2
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Selumetinib 1 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
4
3
2
1
MEKi vs Staurosporine

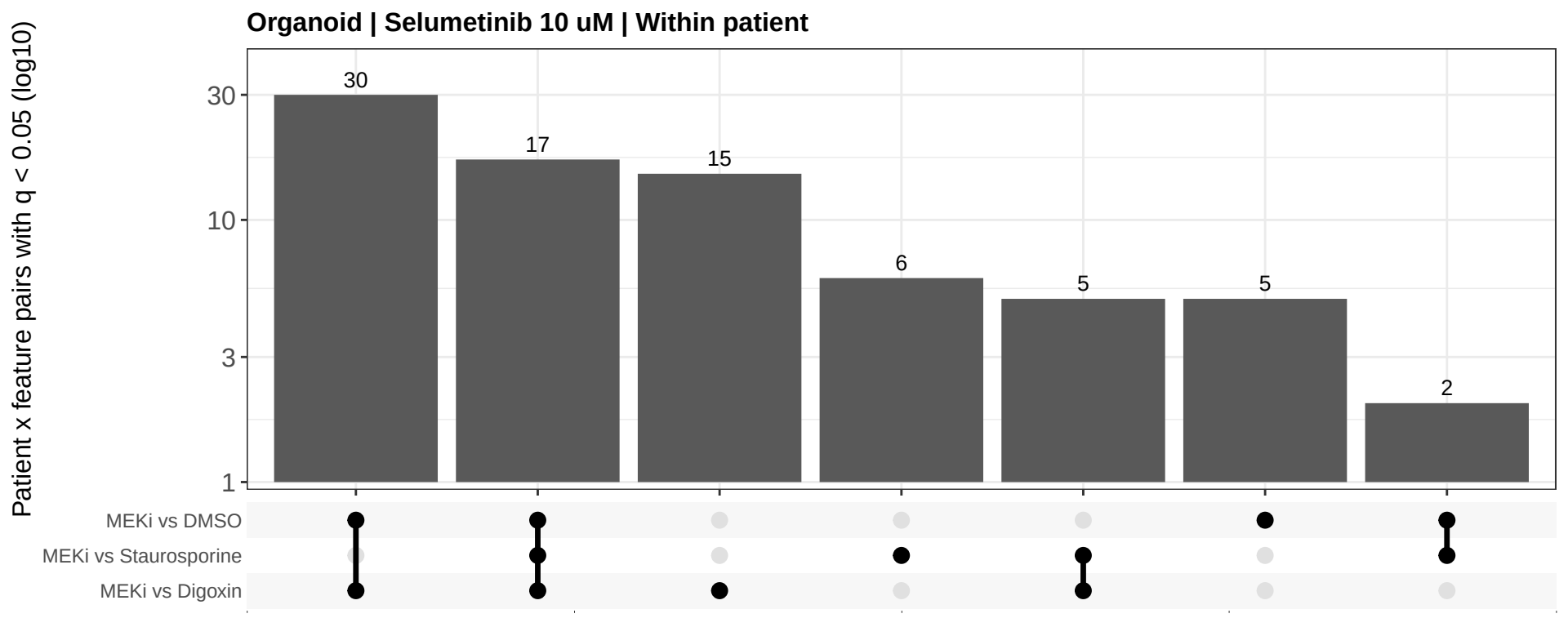

Organoid | Selumetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
10
3
1
8
MEKi vs Staurosporine

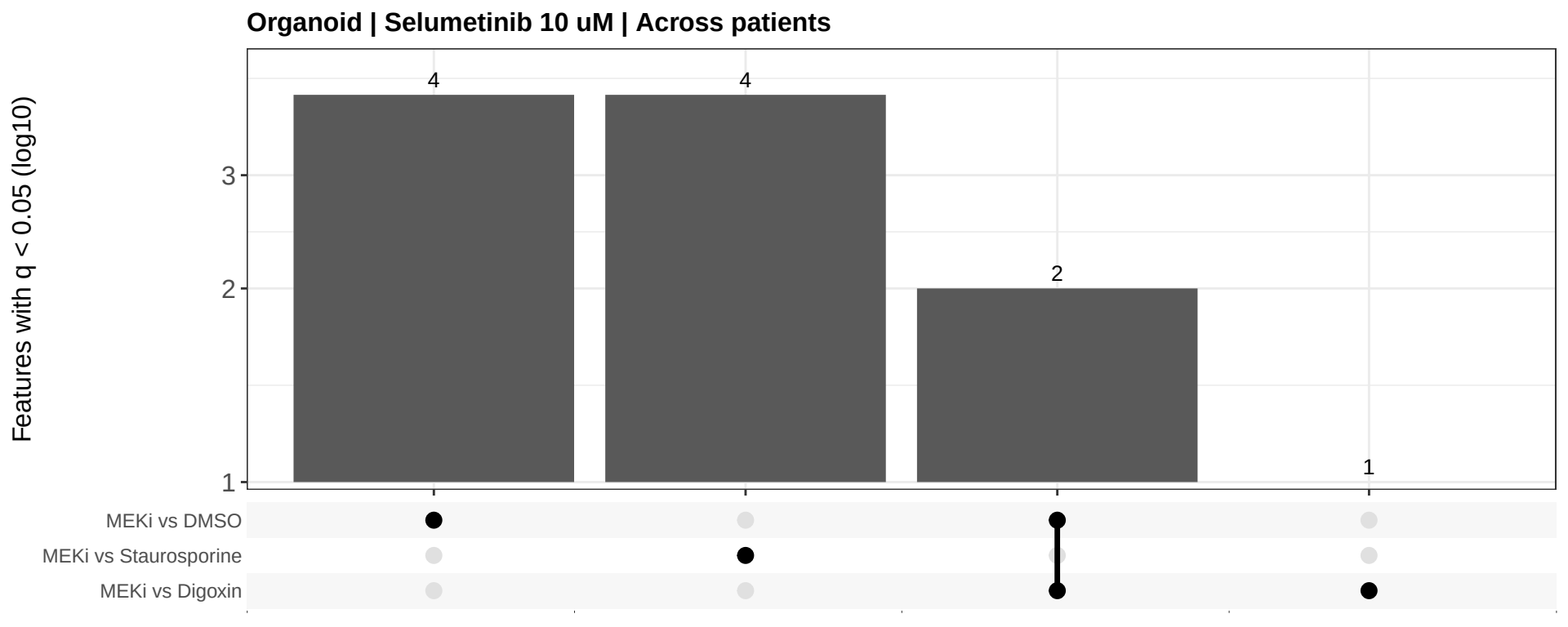
Organoid | Selumetinib 1 uM | Across tumor types (shared)

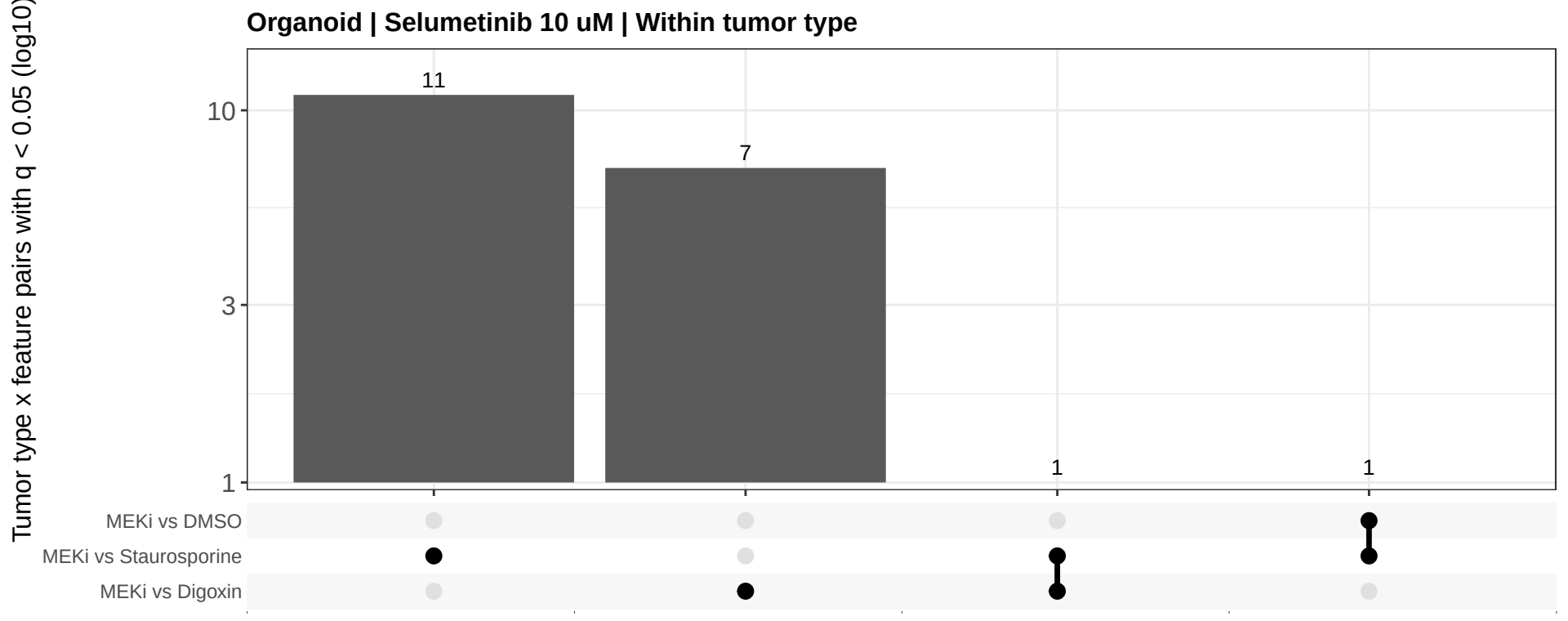
No features with q < 0.05

Organoid | Selumetinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Organoid | Selumetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
30
17
15
6
5
5
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Selumetinib 10 uM | Across patients
Features with q < 0.05 (log10)
4
4
2
1
3
2
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

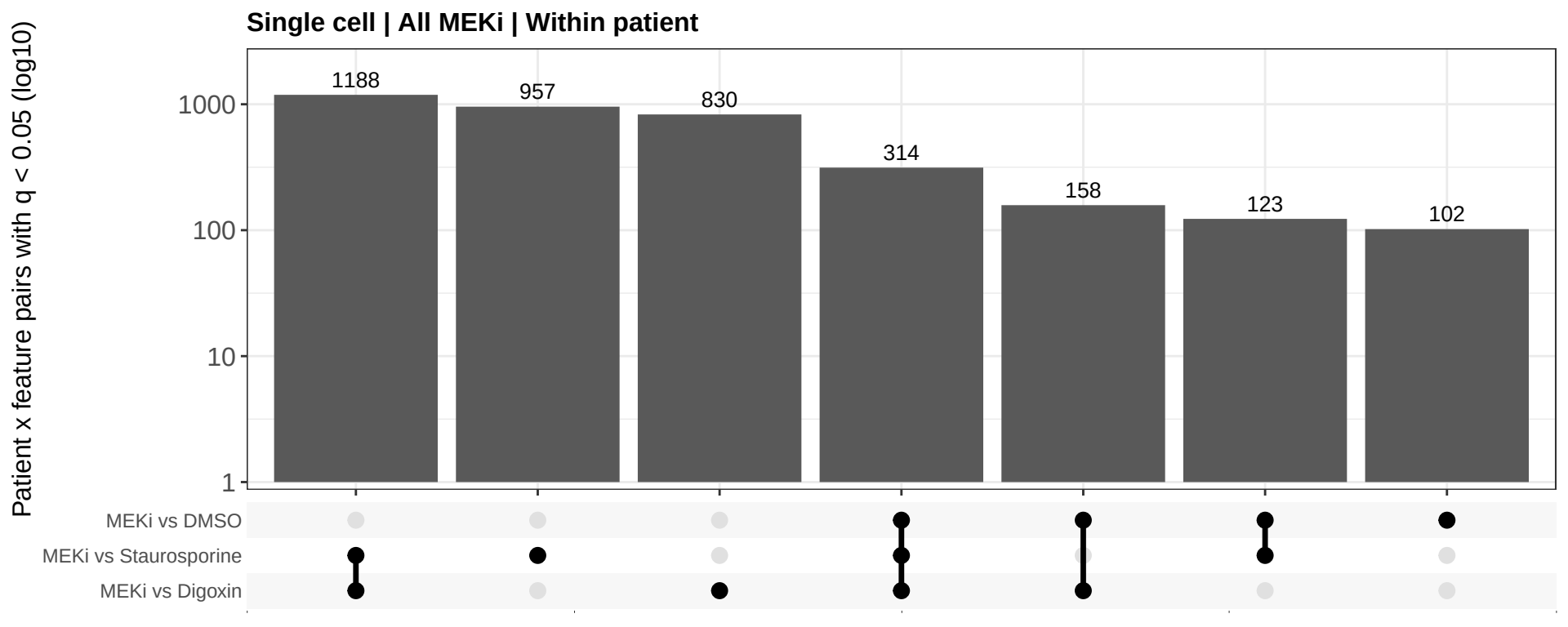

Organoid | Selumetinib 10 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
10
3
1
11
7
1
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

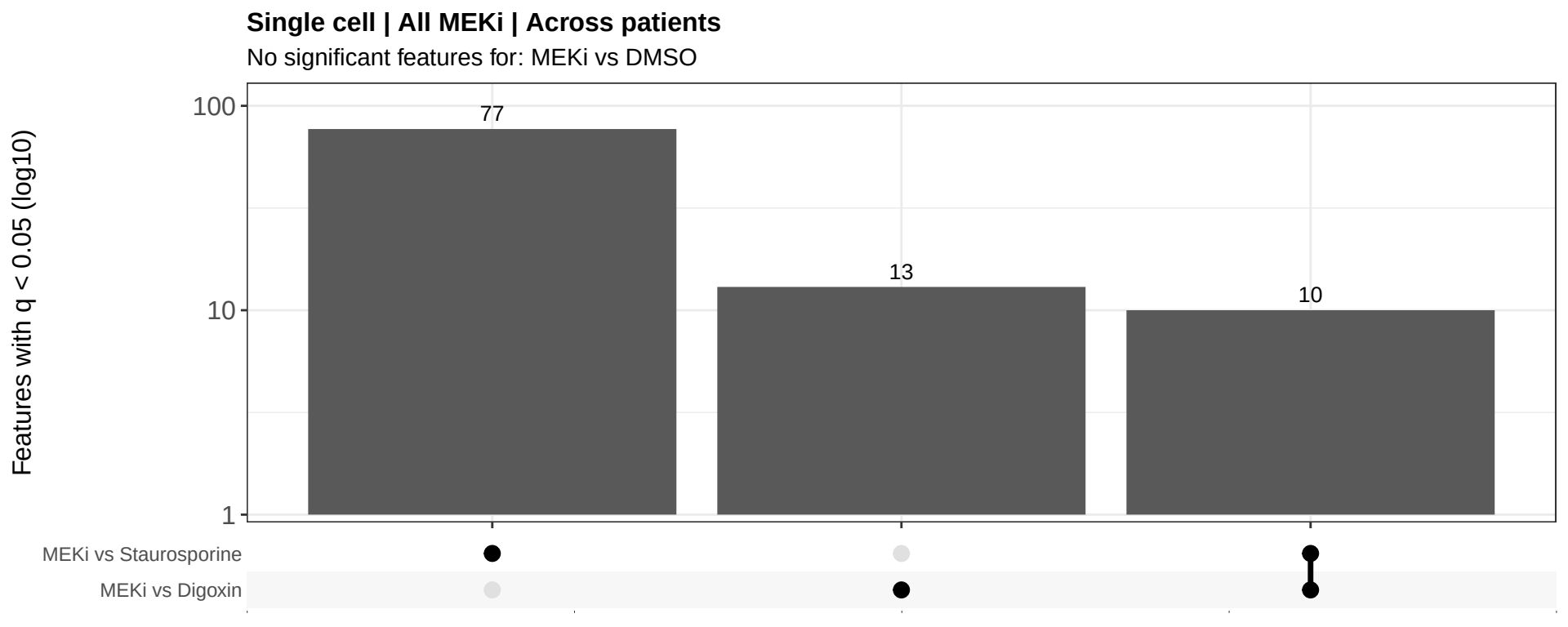
Organoid | Selumetinib 10 uM | Across tumor types (shared)

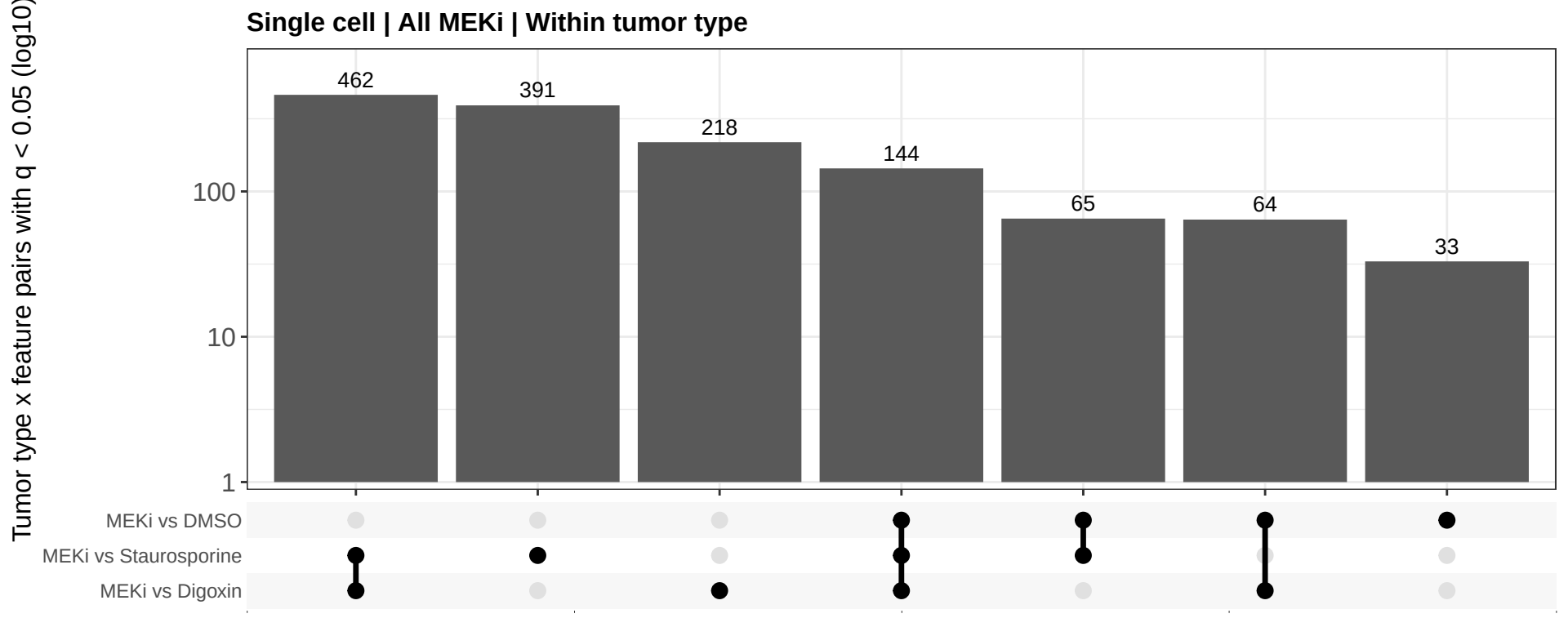
No features with q < 0.05

Organoid | Selumetinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | All MEKi | Within patient
Patient x feature pairs with q < 0.05 (log10)
1000
100
10
1
1188
957
830
314
158
123
102
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | All MEKi | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
100
10
1
77
13
10
MEKi vs Staurosporine
MEKi vs Digoxin

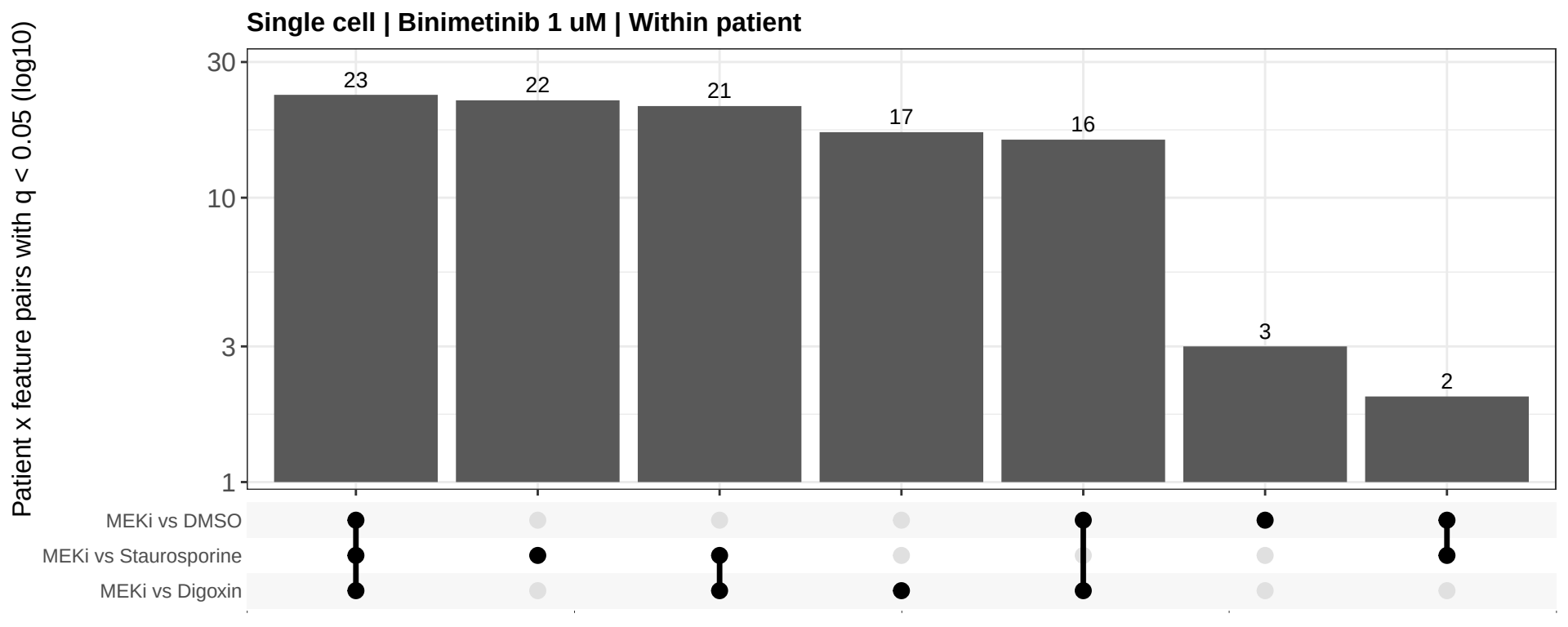

Single cell | All MEKi | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
462
391
218
144
65
64
33
100
10
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | All MEKi | Across tumor types (shared)

No features with q < 0.05

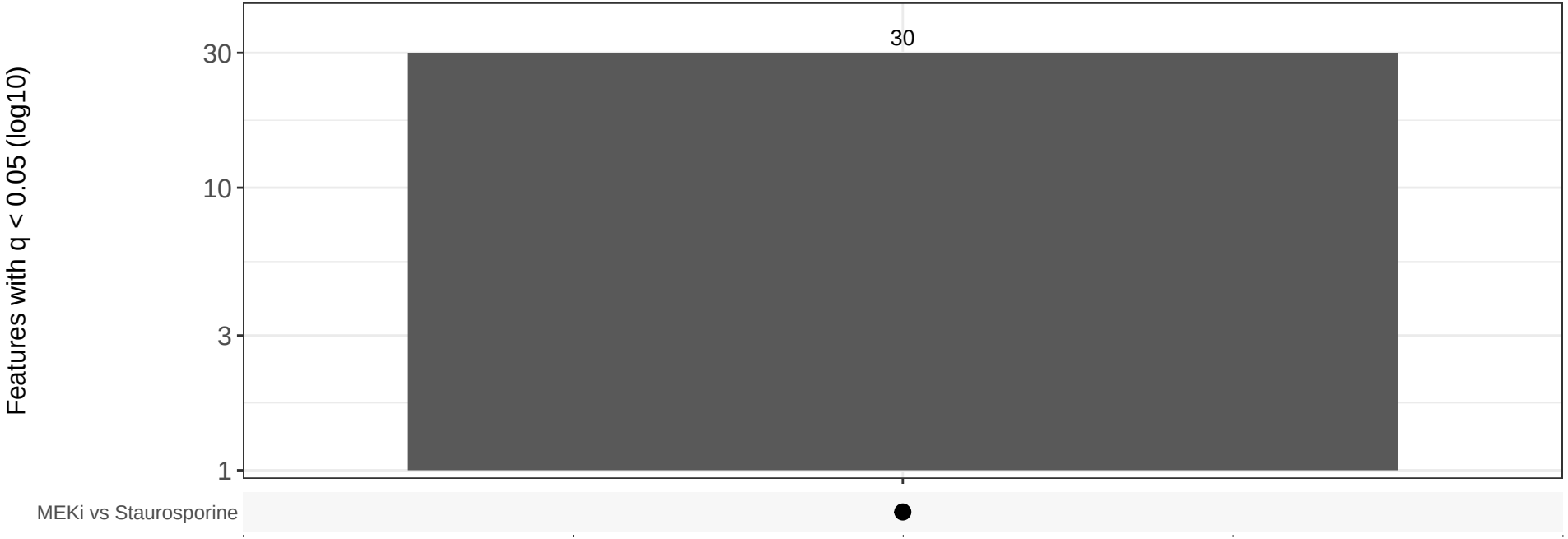
Single cell | All MEKi | Across tumor types (differ)

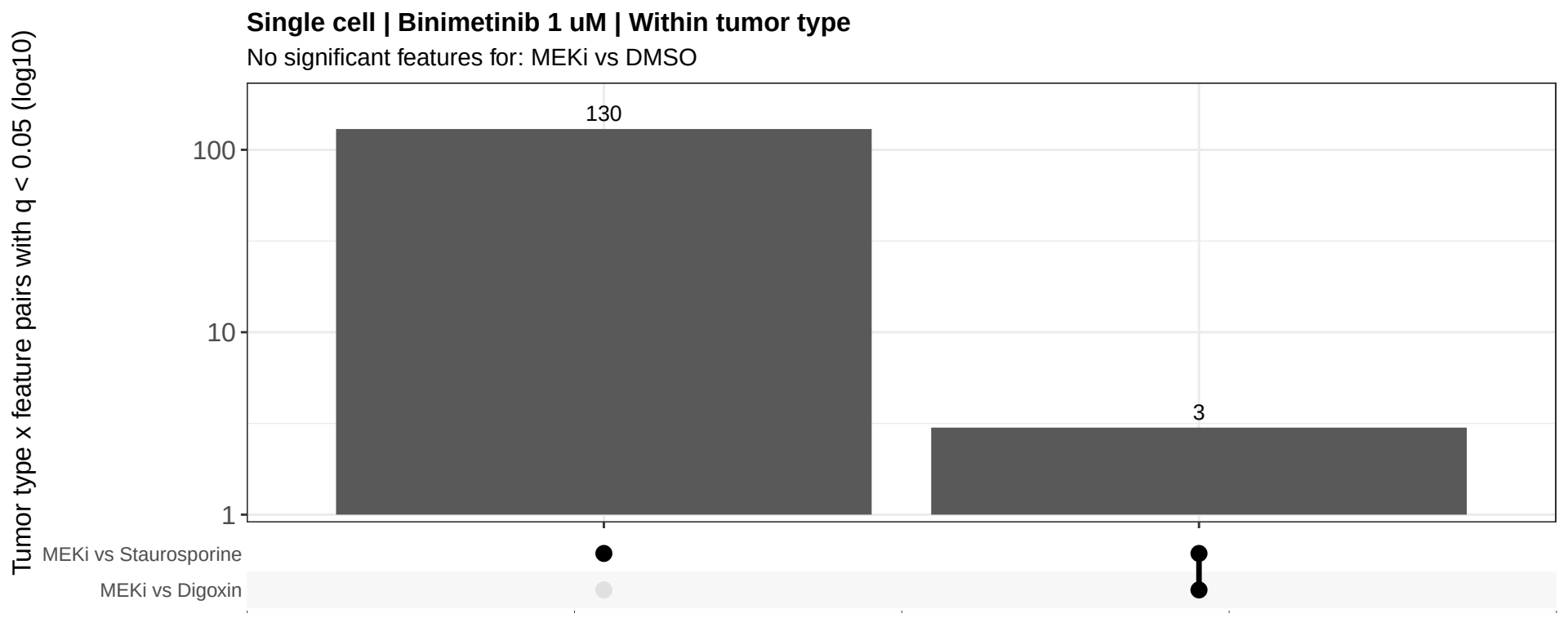
No features with q < 0.05


Single cell | Binimetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
23
22
21
17
16
3
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Binimetinib 1 uM | Across patients

No significant features for: MEKi vs DMSO, MEKi vs Digoxin



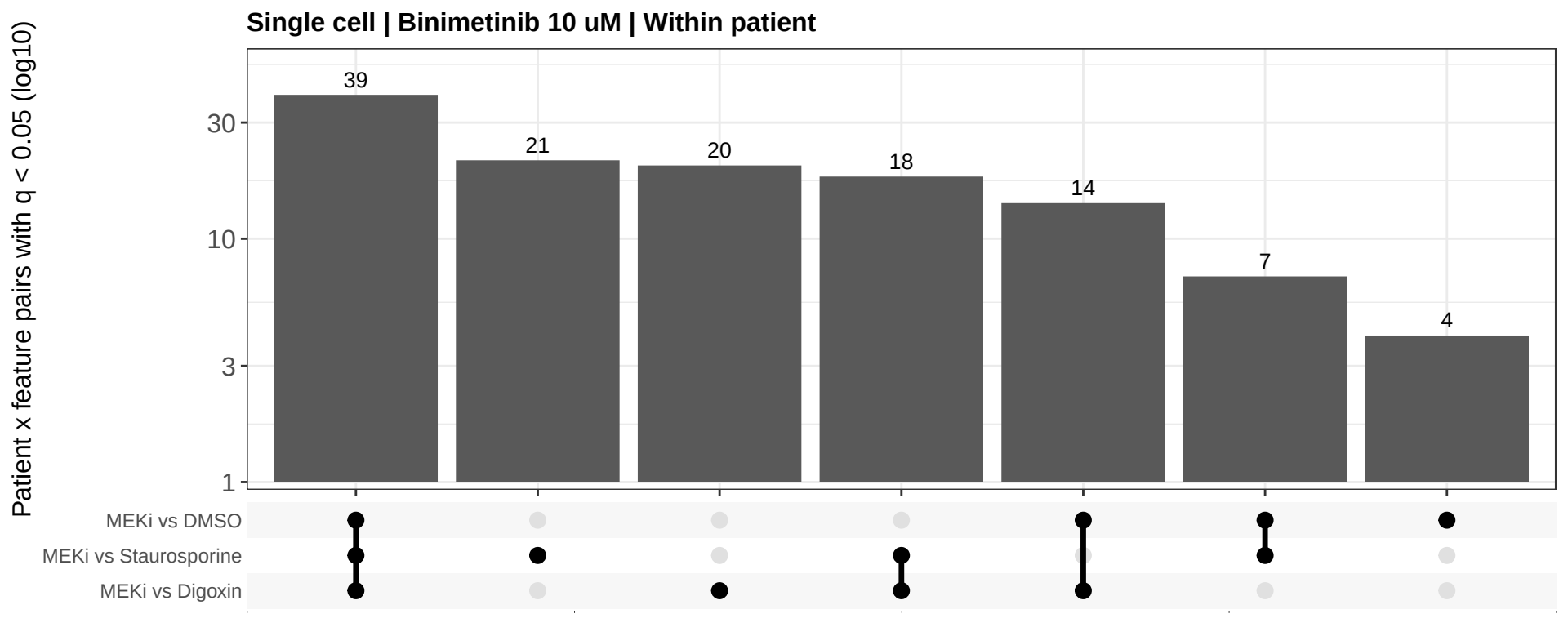

Single cell | Binimetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
130
3
100
10
1
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Binimetinib 1 uM | Across tumor types (shared)

No features with q < 0.05

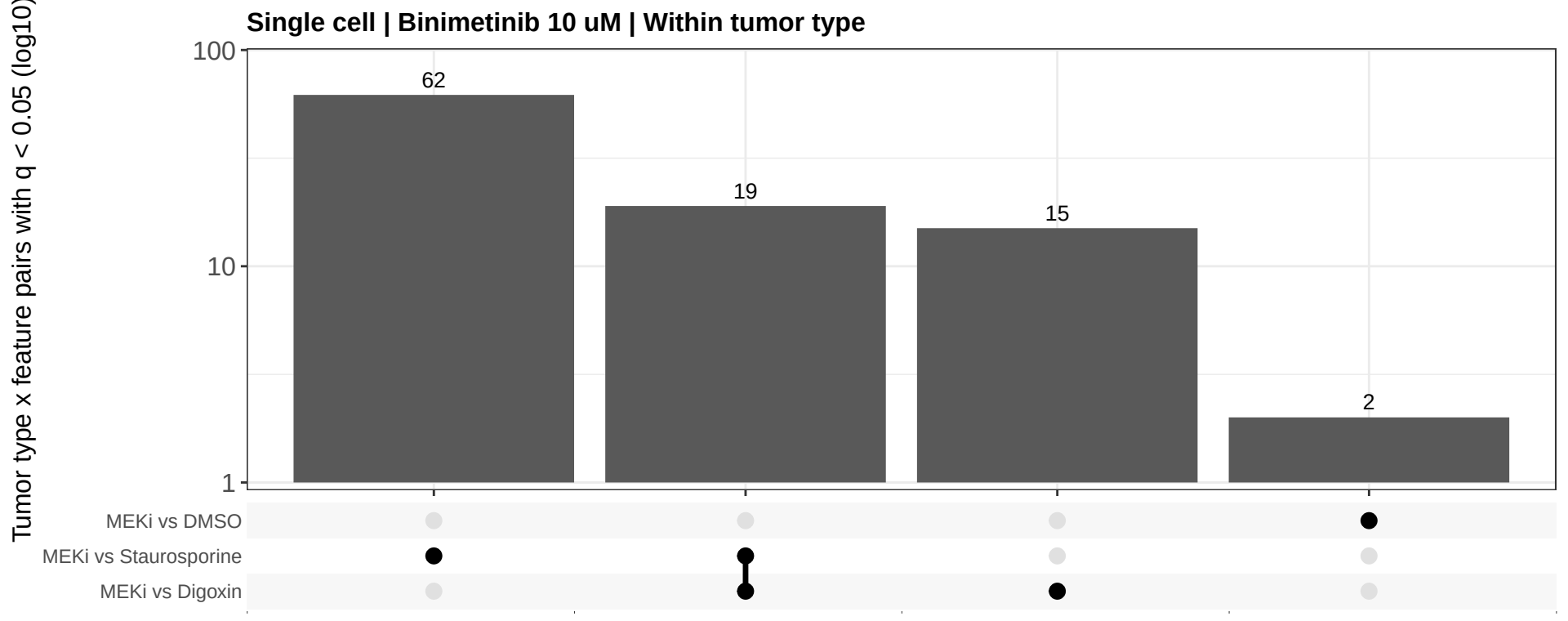
Single cell | Binimetinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Binimetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
39
21
20
18
14
7
4
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Binimetinib 10 uM | Across patients

No features with q < 0.05

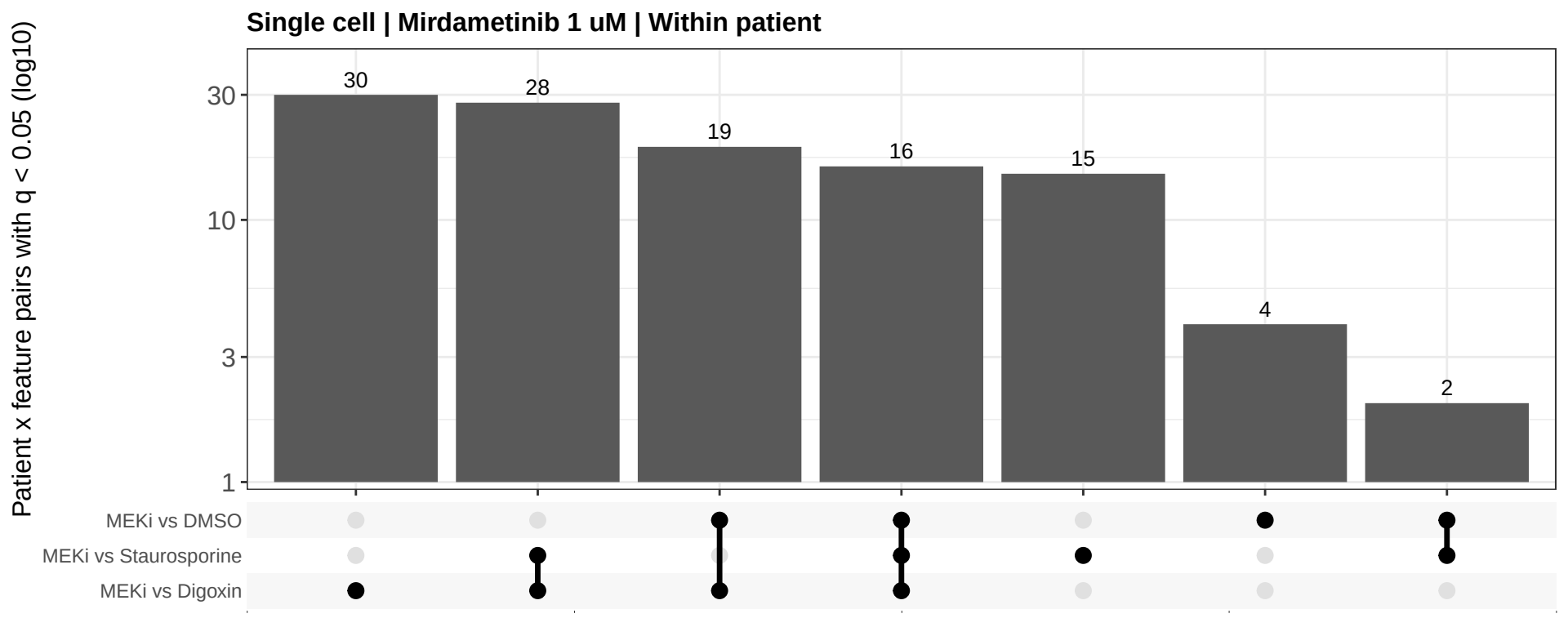

Single cell | Binimetinib 10 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
62
19
15
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

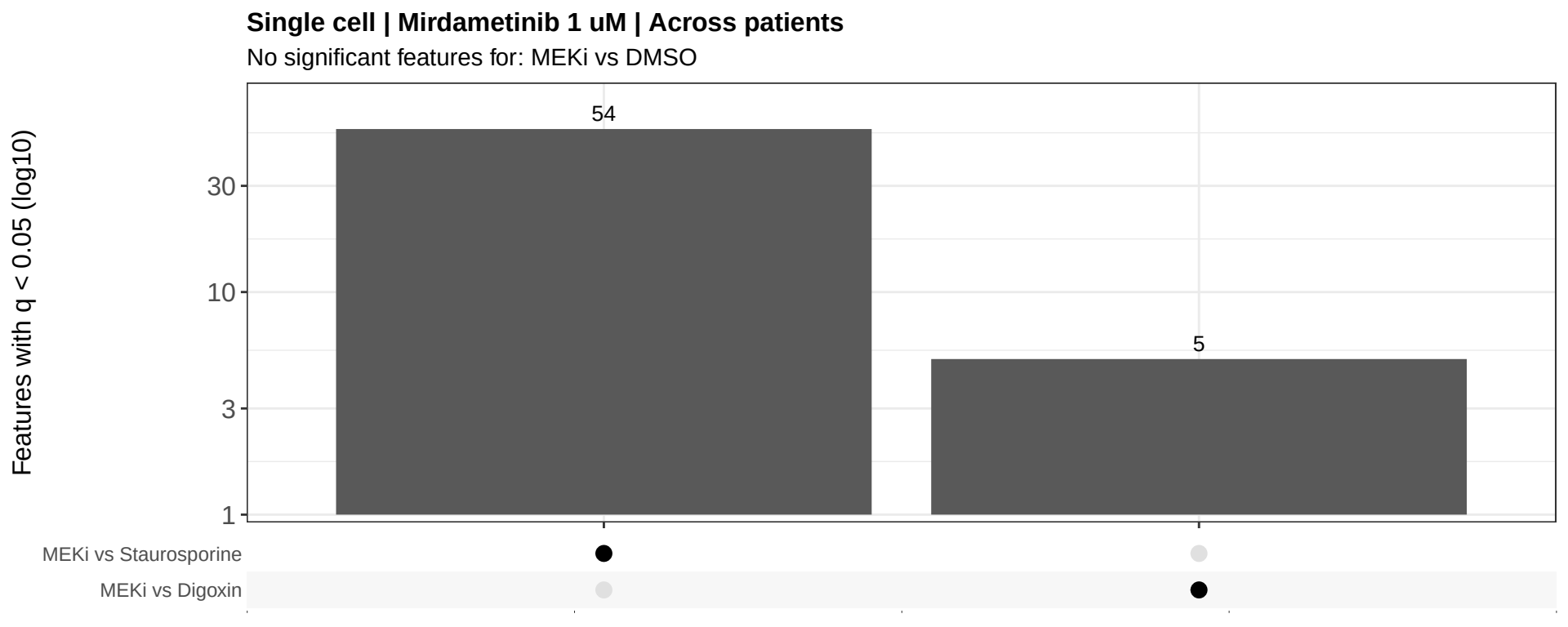
Single cell | Binimetinib 10 uM | Across tumor types (shared)

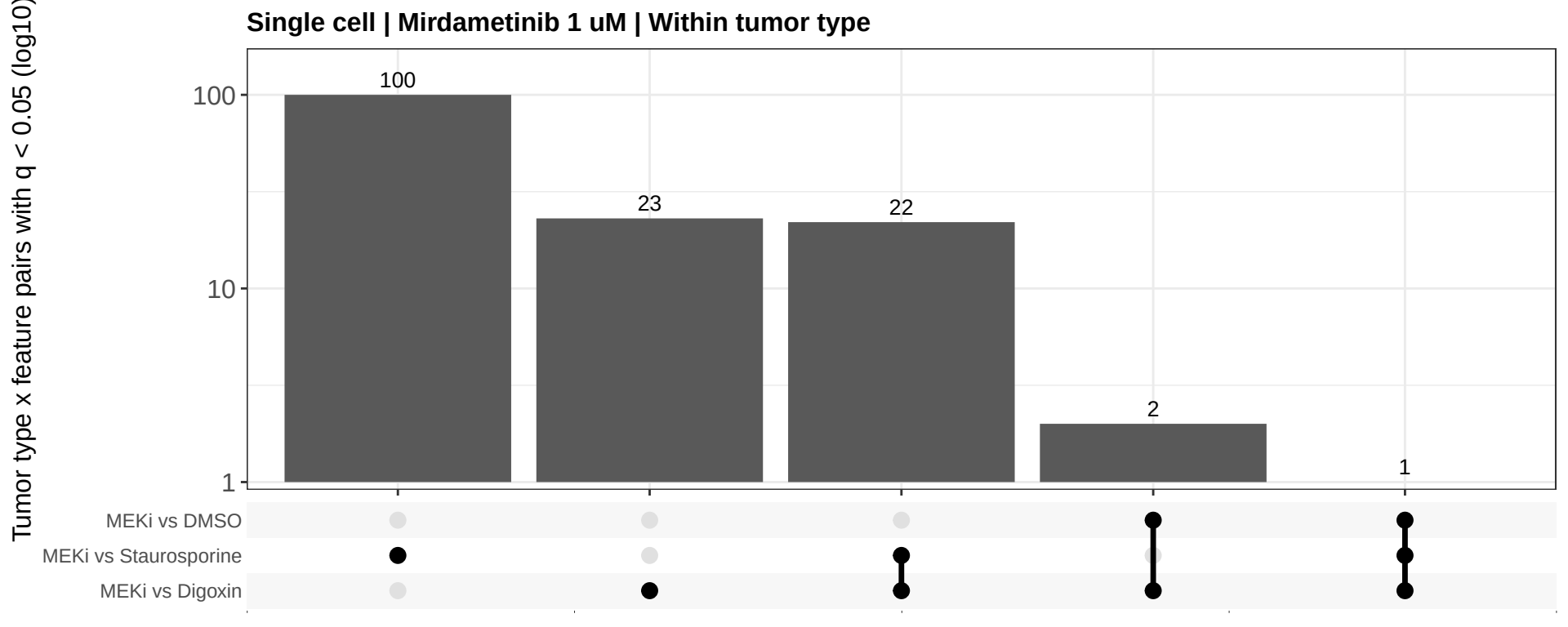
No features with q < 0.05

Single cell | Binimetinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Mirdametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
30
28
19
16
15
4
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Mirdametinib 1 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
54
5
30
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

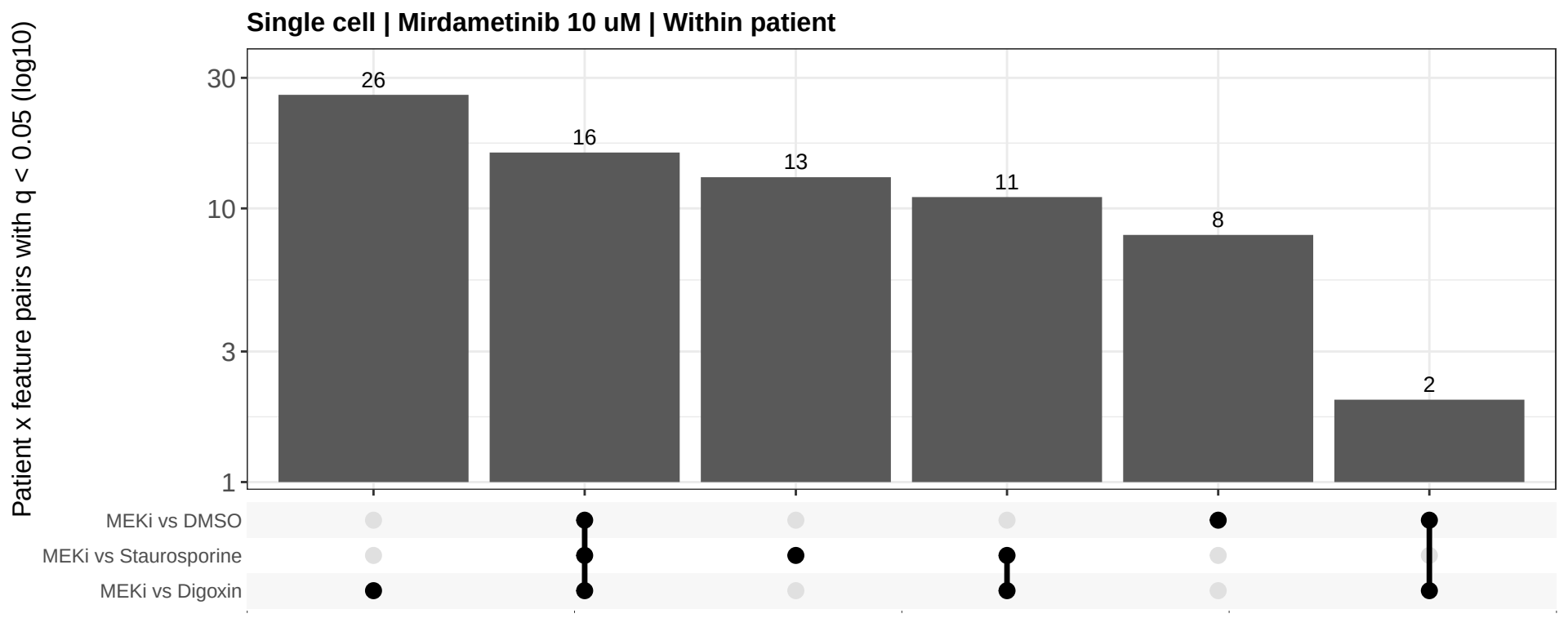

Single cell | Mirdametinib 1 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
100
23
22
2
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

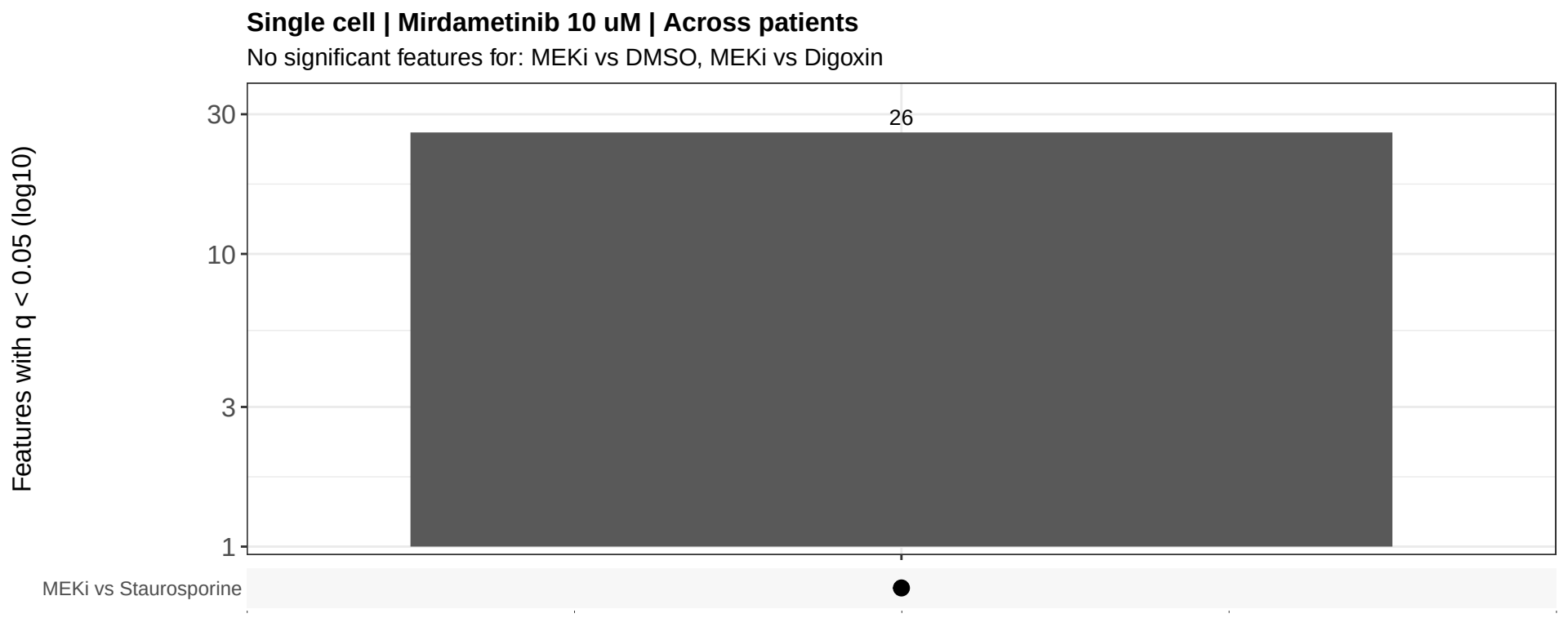
Single cell | Mirdametinib 1 uM | Across tumor types (shared)

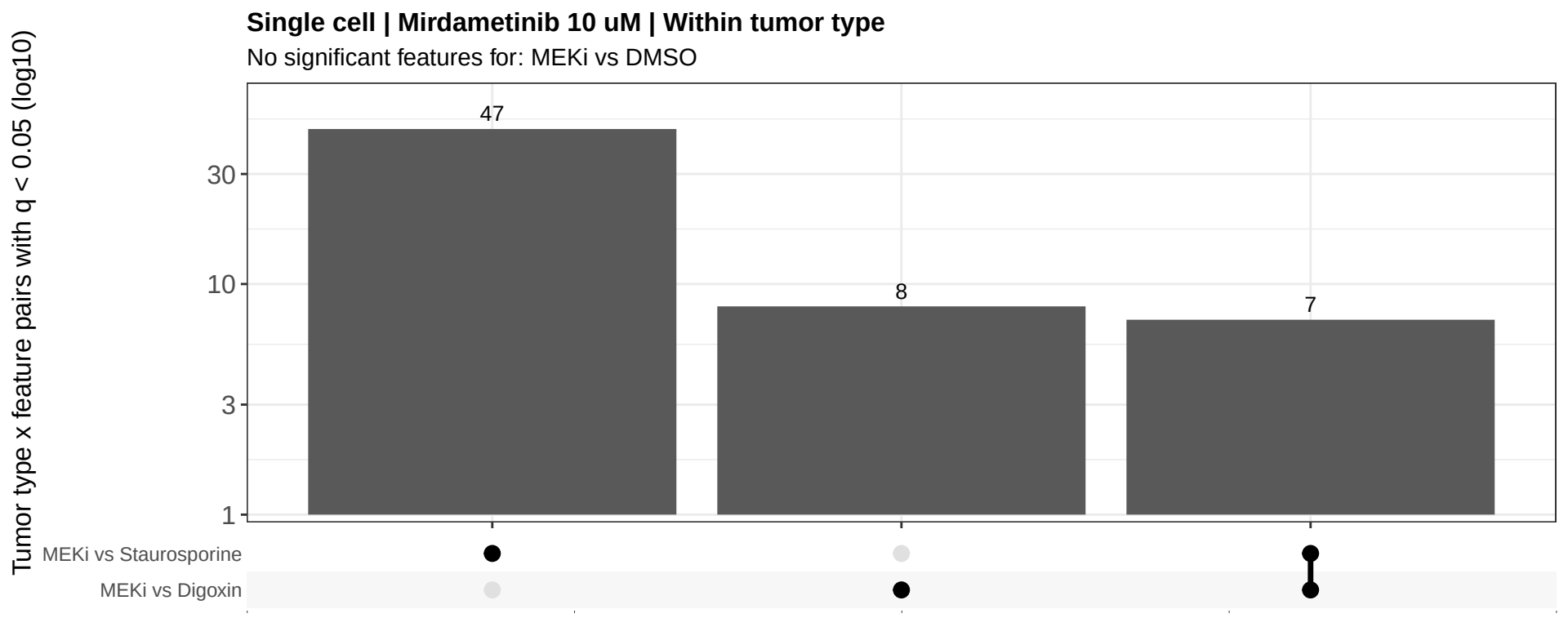
No features with q < 0.05

Single cell | Mirdametinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Mirdametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
26
16
13
11
8
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Mirdametinib 10 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
30
10
3
1
26
MEKi vs Staurosporine

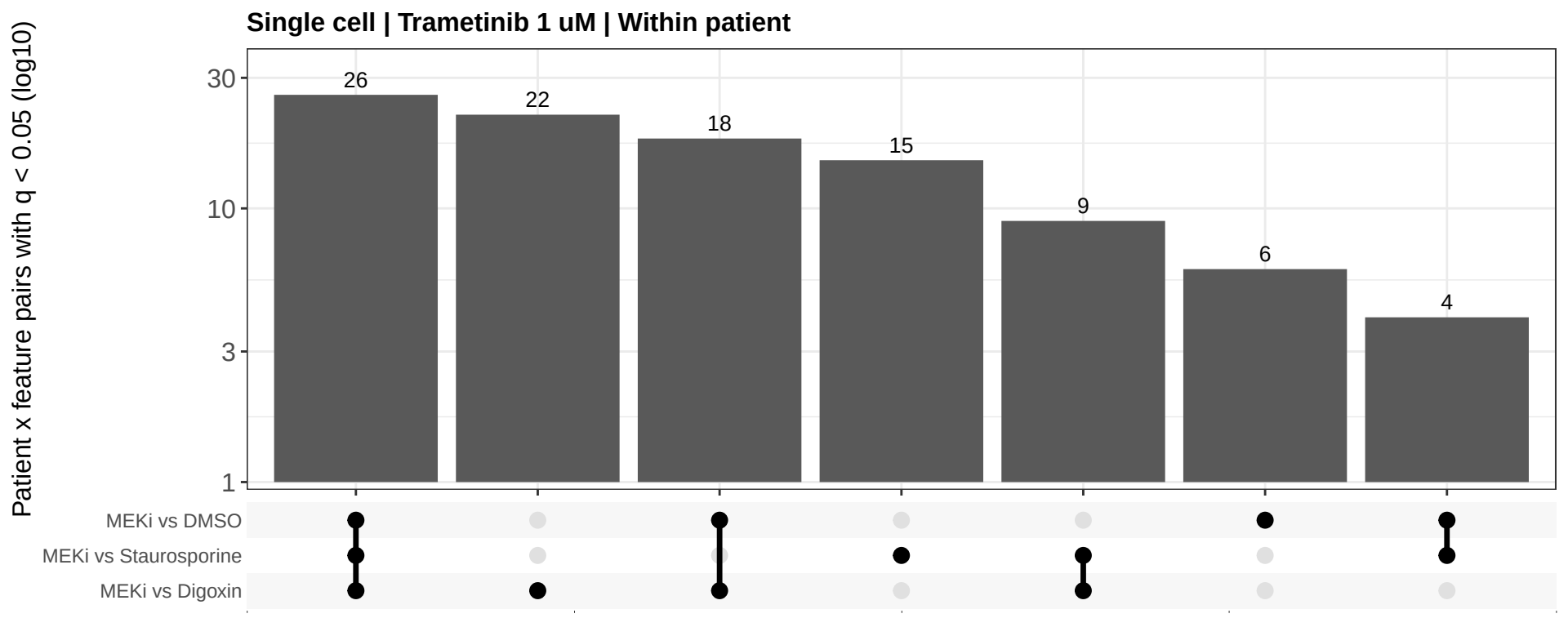

Single cell | Mirdametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
47
8
7
30
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

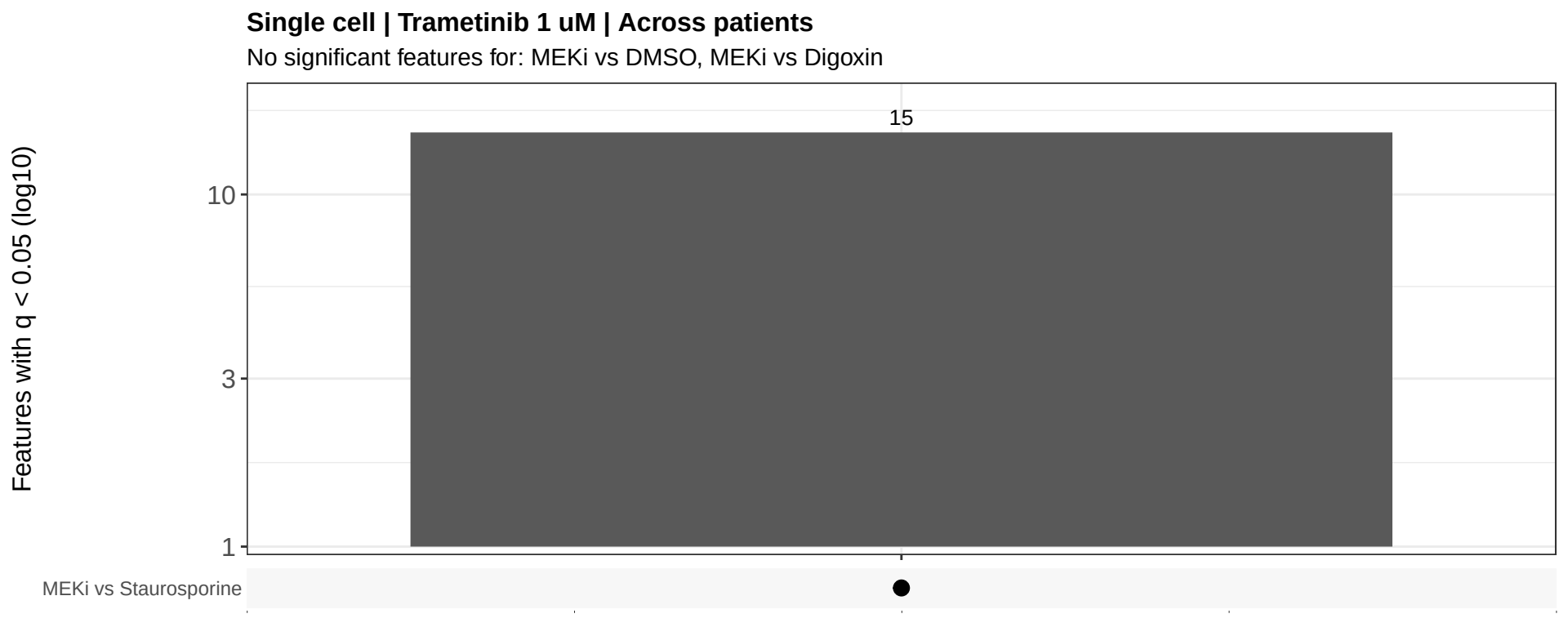
Single cell | Mirdametinib 10 uM | Across tumor types (shared)

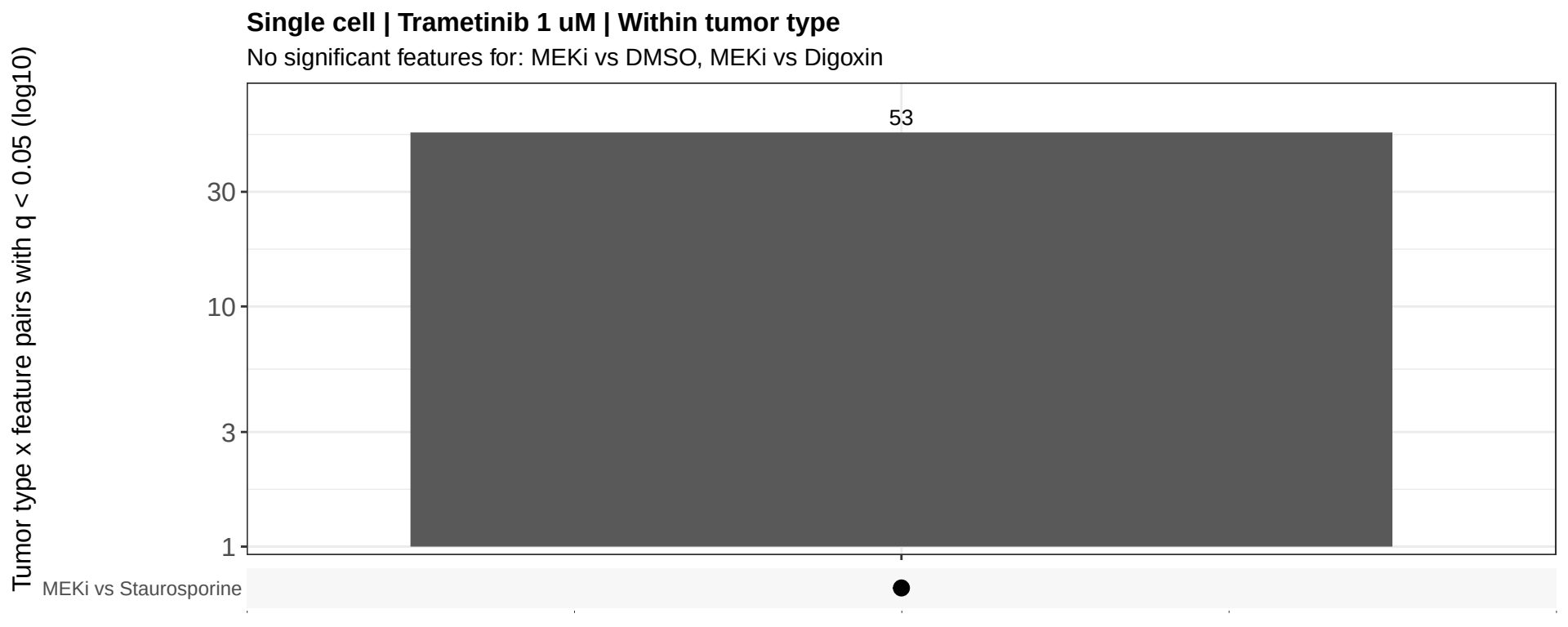
No features with q < 0.05

Single cell | Mirdametinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Trametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
26
22
18
15
9
6
4
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Trametinib 1 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
15
10
3
1
MEKi vs Staurosporine

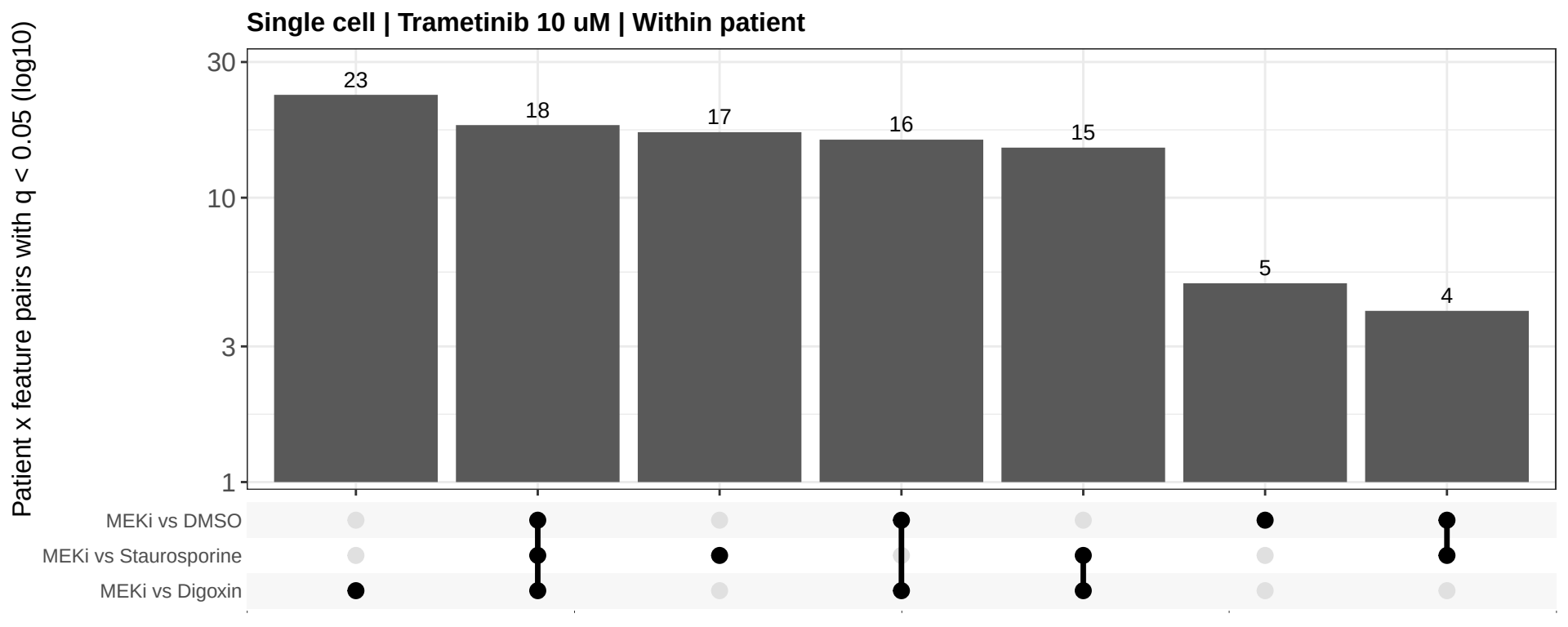

Single cell | Trametinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
53
30
10
3
1
MEKi vs Staurosporine

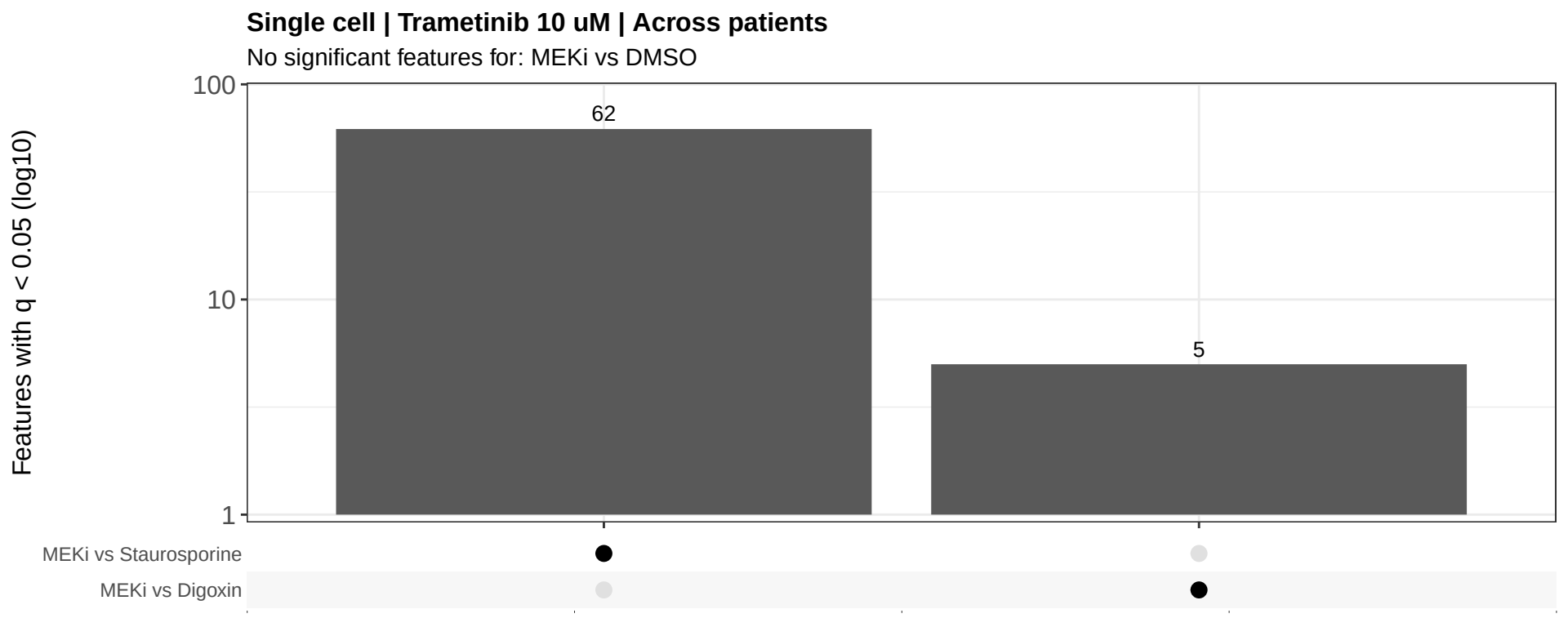
Single cell | Trametinib 1 uM | Across tumor types (shared)

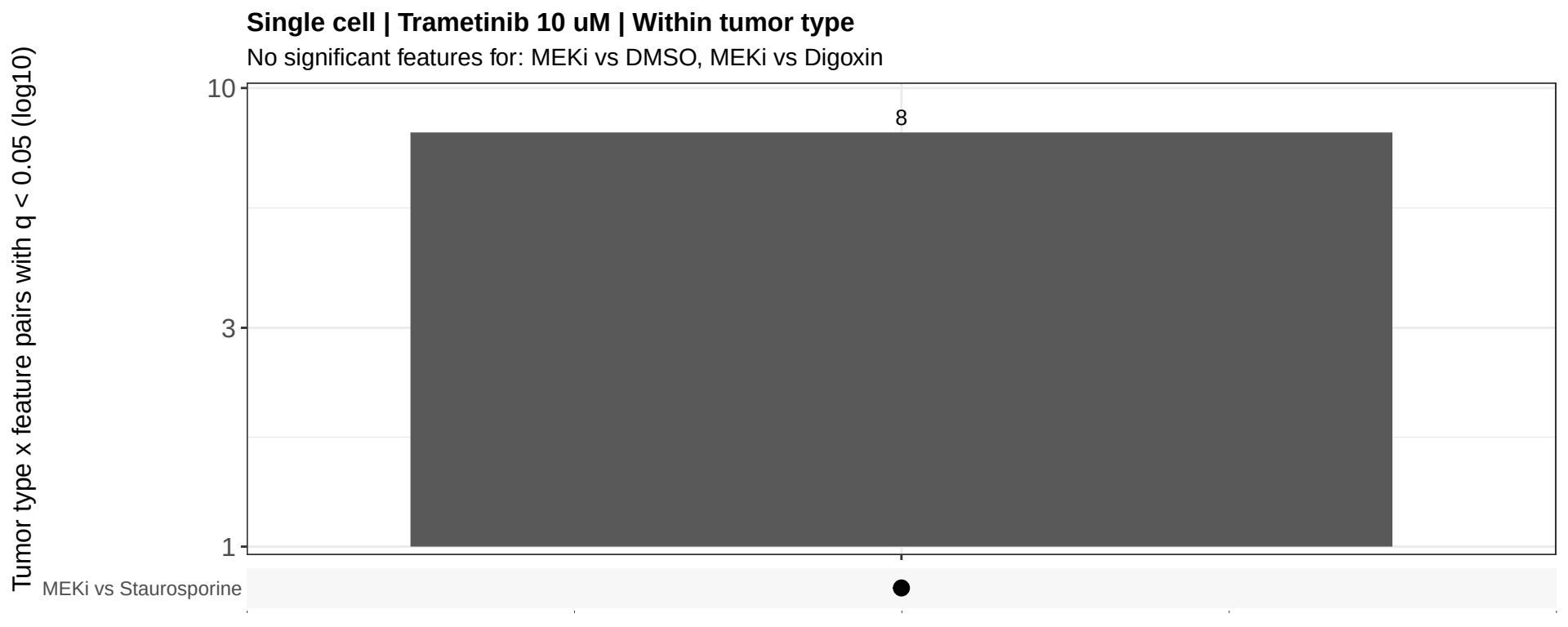
No features with q < 0.05

Single cell | Trametinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Trametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
23
18
17
16
15
5
4
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Trametinib 10 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
100
10
1
62
5
MEKi vs Staurosporine
MEKi vs Digoxin

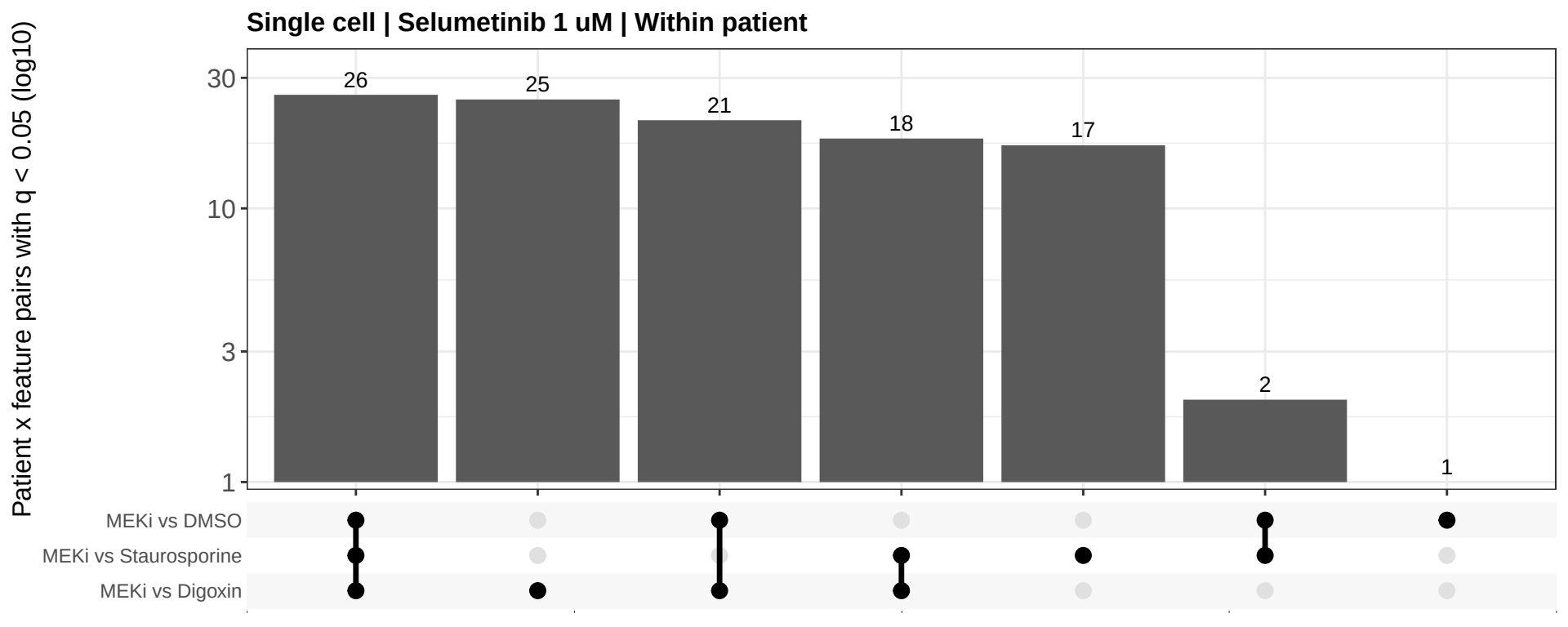

Single cell | Trametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
10
3
1
8
MEKi vs Staurosporine

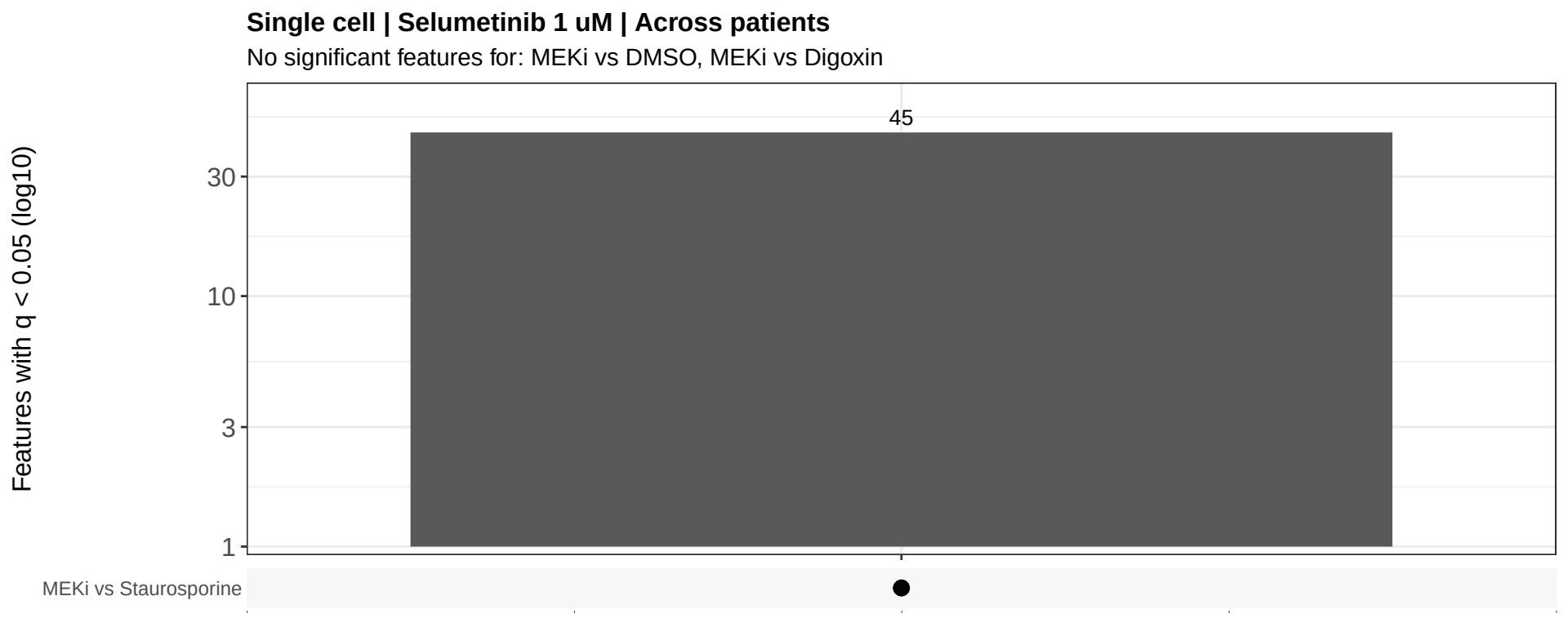
Single cell | Trametinib 10 uM | Across tumor types (shared)

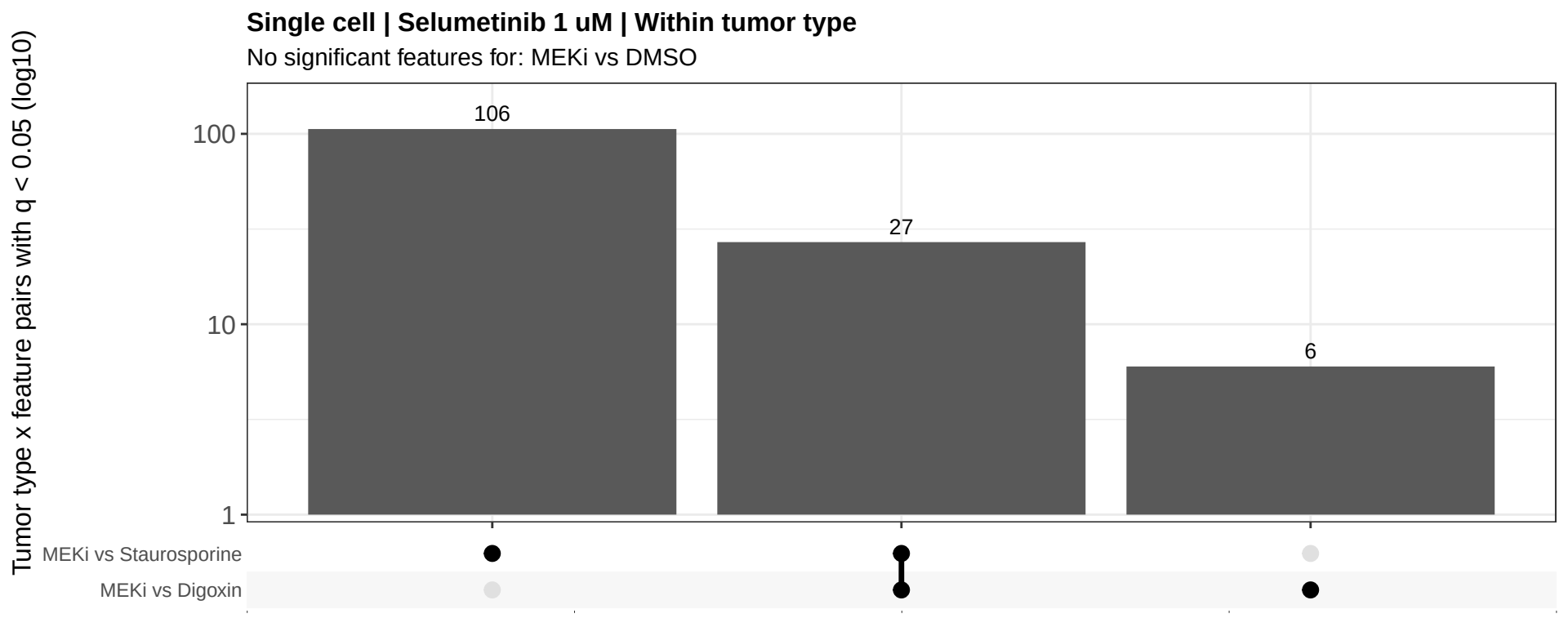
No features with q < 0.05

Single cell | Trametinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Selumetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
26
25
21
18
17
2
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Selumetinib 1 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
45
30
10
3
1
MEKi vs Staurosporine

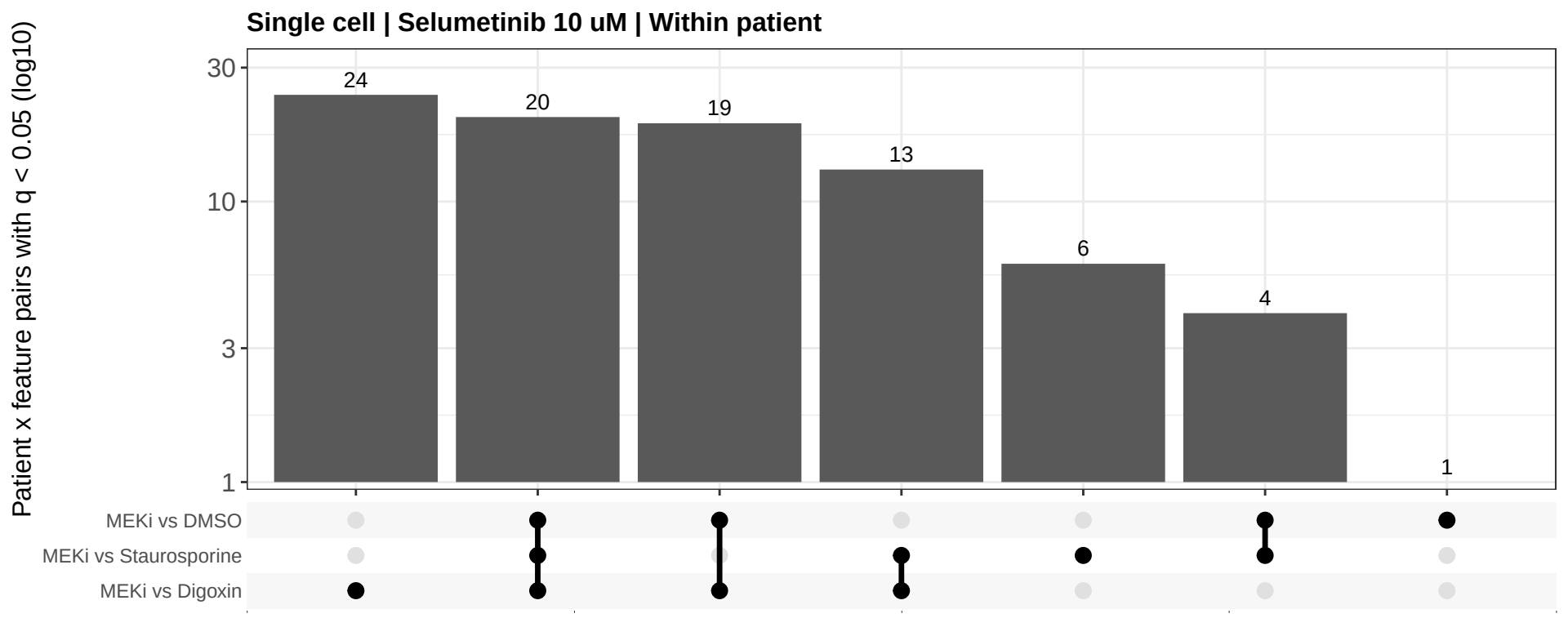

Single cell | Selumetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
106
27
6
MEKi vs Staurosporine
MEKi vs Digoxin

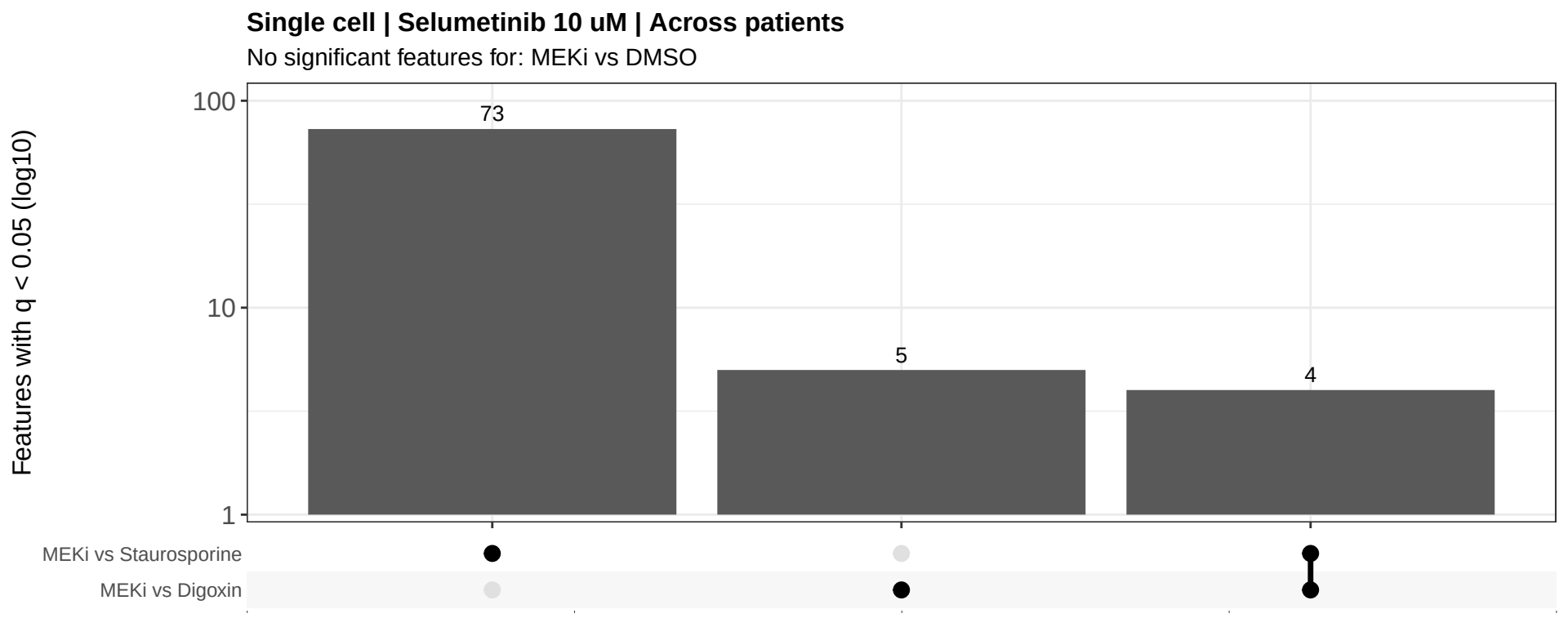
Single cell | Selumetinib 1 uM | Across tumor types (shared)

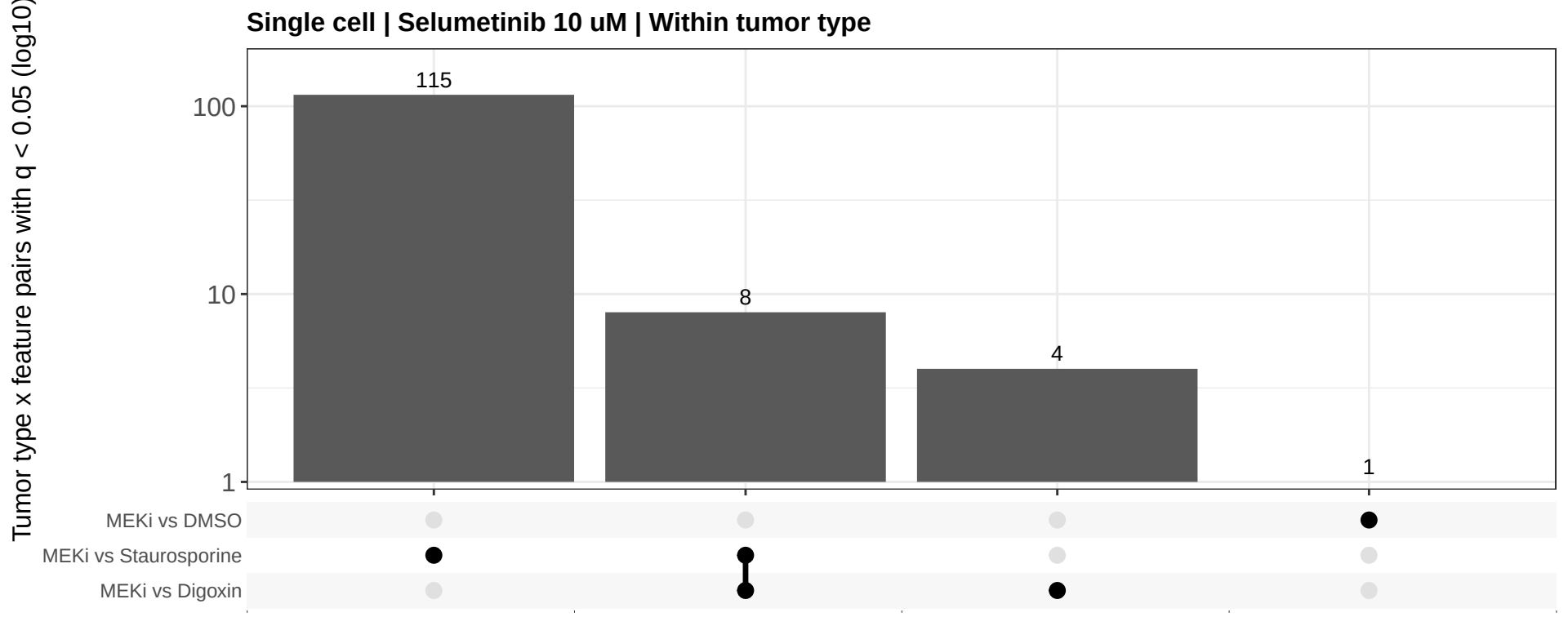
No features with q < 0.05

Single cell | Selumetinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Selumetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
24
20
19
13
6
4
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Selumetinib 10 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
100
10
1
73
5
4
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Selumetinib 10 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
115
8
4
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Selumetinib 10 uM | Across tumor types (shared)

No features with q < 0.05

Single cell | Selumetinib 10 uM | Across tumor types (differ)

No features with q < 0.05